GRADUATION THESIS

QUADROTOR MODELLING CONTROL AND IMPLEMENTATION ON REAL SYSTEM



CONTROL AND AUTOMATION ENGINEERING

NECİP BAHA BAĞRIAÇIK

TUNAHAN EMİR KARABAY

AGENDA

- OBJECTIVES OF THE STUDY
- INTRODUCTION OF QUADROTOR
- MODELLING OF SYSYEM
  - Quadrotor Dynamics
  - Kinematic Model
  - Quadrotor Nonlinear Dynamic
  - Linear Model of Quadrotor (Linearization)
  - State-Space Representation
  - BLDC Motor System Identification
- DESIGN AND BUILD
  - Mechanical Design
  - Hardware Component and Integration
- CONTROL METHOD
  - Controller Design with Simulink
  - Tuning of the Controller Coefficient
- IMPLEMENTATION OF HARDWARE
  - Pixhawk Calibrations
  - Firmware Deploy
- RESULTS AND DISCUSSIONS

OBJECTIVES OF THE STUDY

- To provide new information to the aviation literature, specifically the quadrotor.
- To develop quadrotors by conducting research in the field of mechanics.
- Providing innovative solutions in the field of hardware.
- Increasing the functionality of the quadrotor with software improvements.
- Optimizing the performance of the quadrotor by working on control systems.

INTRODUCTION OF QUADROTOR

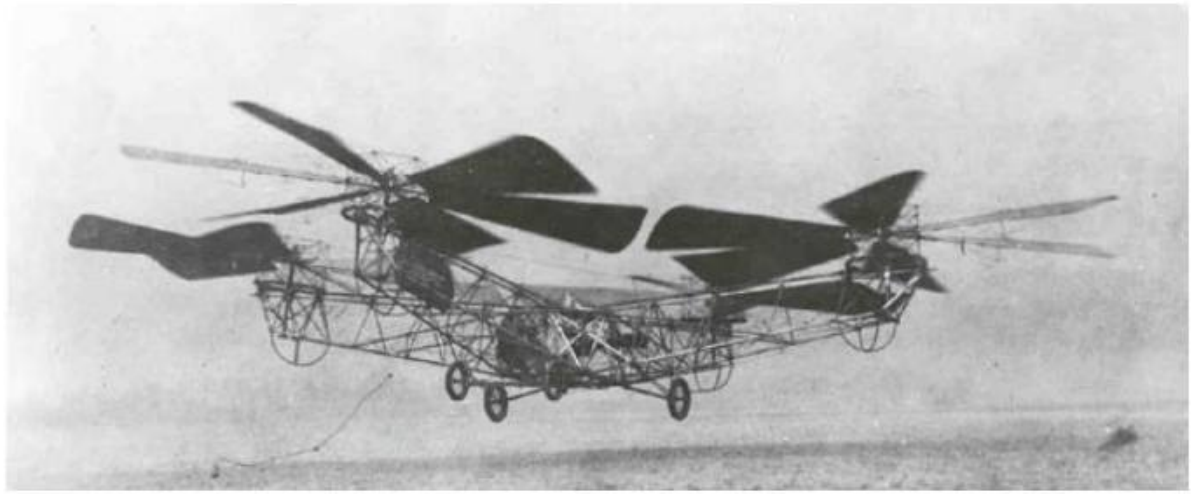


Figure 1:The first quadrotor image in history

Quadrotors are unmanned aerial vehicles with four propellers, offering high maneuverability and efficiency. This study focuses on the design and implementation of controllers for the construction and stable flight of quadrotors. By addressing the challenges in control system development, it aims to enhance the performance and reliability of quadrotor operations.

MODELLING OF SYSYEM

Quadrotor Dynamics

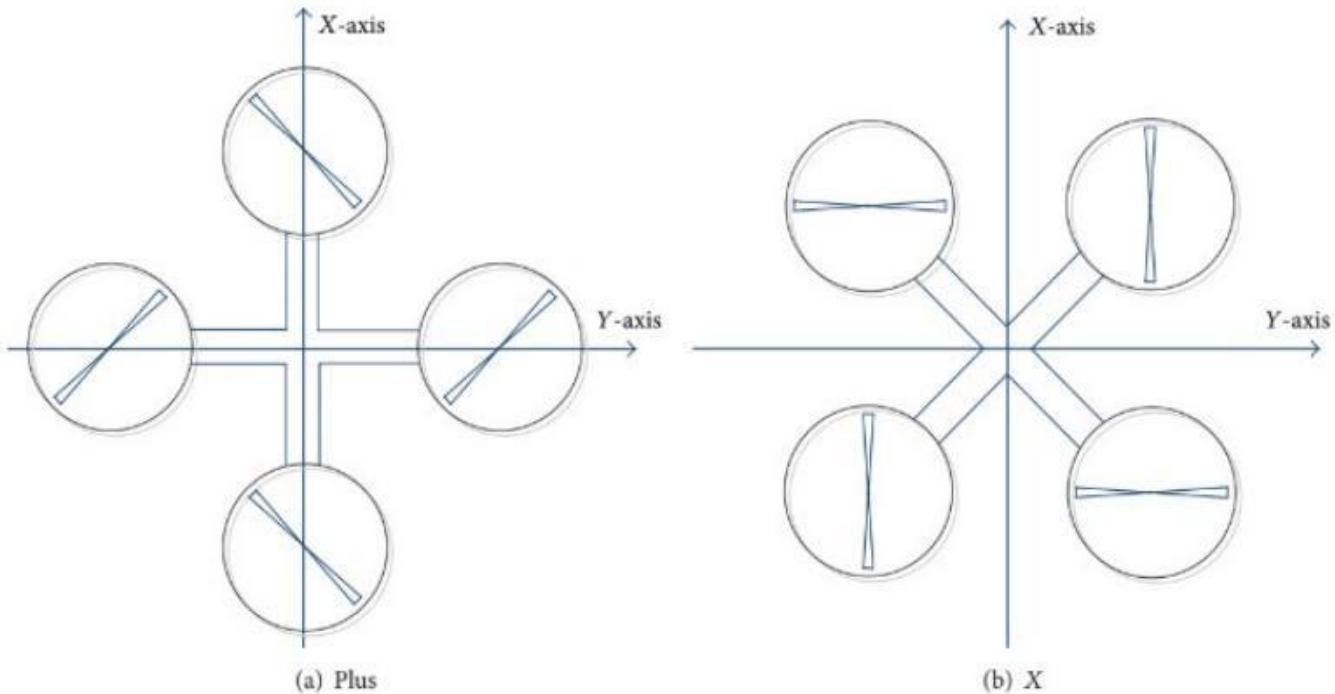


Figure 2: Images of quadrotor types in the shapes of plus and X.

$$F_i = \frac{1}{2}\rho A b\, r^2 \Omega_i^2$$

$$M_i = \frac{1}{2}\rho A d\, r^2 \Omega_i^2$$

$$f_t = U_1 = b(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2)$$

$$T_x = U_2 = bl(-\omega_1^2 - \omega_2^2 + \omega_3^2 + \omega_4^2)$$

$$T_y = U_3 = bl(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2)$$

$$T_z = U_4 = d(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$$

MODELLING OF SYSYEM

Kinematic Model

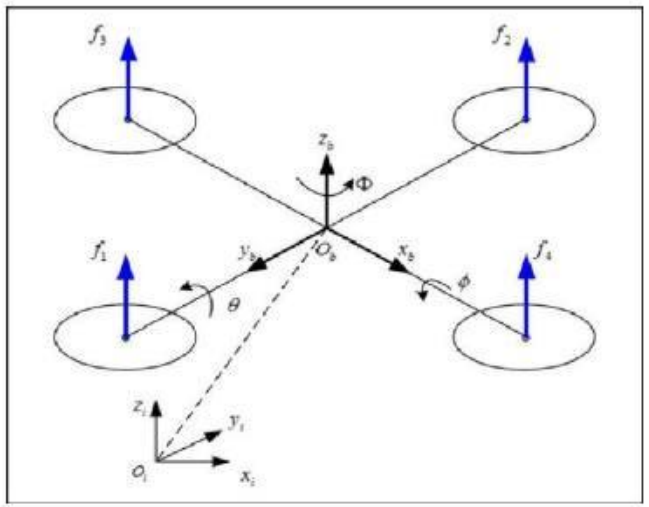


Figure 3: Quadrotor Axis.

$$R = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \sin\psi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\cos\psi + \sin\phi\sin\theta\sin\psi & \sin\phi\cos\theta \\ \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi & \sin\theta\cos\phi\sin\psi - \sin\phi\cos\psi & \cos\theta\cos\phi \end{bmatrix}$$

$$\boldsymbol{v} = \boldsymbol{R}.\boldsymbol{v_B}$$

$$\boldsymbol{\omega} = \boldsymbol{T}.\boldsymbol{\omega_B}$$

Where $\boldsymbol{v} = [\dot{x} \quad \dot{y} \quad \dot{z}]^T$, $\boldsymbol{\omega} = [\dot{\phi} \quad \dot{\theta} \quad \dot{\psi}]^T$, $\boldsymbol{v_B} = [u \quad v \quad \omega]^T$, $\boldsymbol{\omega_B} = [p \quad q \quad r]^T$ and

T is a matrix for angular transformations.

$$T = \begin{bmatrix} 1 & \sin\phi\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \dfrac{\sin(\phi)}{\cos(\theta)} & \dfrac{\cos(\phi)}{\cos(\theta)} \end{bmatrix}$$

MODELLING OF SYSYEM

Quadrotor Nonlinear Dynamic

$$\dot{x} = w[s(\varphi)\, c(\psi) + c(\varphi)\, c(\psi)\, s(\theta)] - v[c(\varphi)\, s(\psi) - c(\psi)\, s(\varphi)\, s(\theta) + u[c(\psi)\, c(\theta)]$$

$$\dot{y} = v[c(\varphi)\, c(\psi) + s(\varphi)\, s(\psi)\, s(\theta)] - w[c(\psi)s(\varphi) - c(\varphi)s(\psi)s(\theta) + u[c(\theta)\, s(\psi)]]$$

$$\dot{z} = w[c(\varphi)c(\theta)] - u[s(\theta)] + v[c(\theta)s(\varphi)]$$

$$\dot{w} = (uq - vp) - g\, c(\theta)\, s(\varphi)\frac{U_1}{m}$$

$$\dot{\phi} = \dot{\phi} + \dot{\psi}[c(\phi)\, t(\theta)] + \dot{\theta}[s(\phi)\, t(\theta)]$$

$$\ddot{\phi} = \frac{I_y - I_z}{I_x}(\dot{\theta}\dot{\psi}) + \frac{U_2}{I_x}$$

MODELLING OF SYSYEM

Quadrotor Nonlinear Dynamic

$$\dot{\theta} = \dot{\theta}[c(\phi)] - \dot{\psi}[s(\phi)]$$

$$\ddot{\theta} = \frac{I_z - I_x}{I_y}(\dot{\phi}\dot{\psi}) + \frac{U_3}{I_y}$$

$$\dot{\psi} = \dot{\psi}\left(\frac{s(\phi)}{c(\theta)}\right) + \dot{\theta}\left(\frac{s(\phi)}{c(\theta)}\right)$$

$$\ddot{\psi} = \frac{I_x - I_y}{I_z}(\dot{\phi}\dot{\theta}) + \frac{U_4}{I_z}$$

$$\dot{u} = (vr - wq) + g\,s(\theta)$$

$$\dot{v} = (wp - ur) - g\,c(\theta)s(\varphi)$$

$$m(\omega_B\, x\, v_B + \dot{v}_B) = f_B$$

MODELLING OF SYSYEM

Linear Model of Quadrotor (Linearization)

$$f(x,u) \cong f(x_0,u_0) + \left.\frac{\partial f}{\partial x}\right|_{x=x_0\, u=u_0} (x - x_0) + \left.\frac{\partial f}{\partial u}\right|_{x=x_0\, u=u_0} (u - u_0) + H.O.T$$

$$\boldsymbol{x}_0 = [x \quad 0 \quad y \quad 0 \quad z \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T$$

$$\boldsymbol{u}_0 = [mg \quad 0 \quad 0 \quad 0]^T$$

MODELLING OF SYSYEM

State-Space Representation

where x is the state vector and is defined as:

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u} \qquad \boldsymbol{x} = [x \quad \dot{x} \quad y \quad \dot{y} \quad z \quad \dot{z} \quad \phi \quad \dot{\phi} \quad \theta \quad \dot{\theta} \quad \psi \quad \dot{\psi}]^T$$

$$\boldsymbol{y} = C\boldsymbol{x} + D\boldsymbol{u} \qquad \boldsymbol{y} = [x \quad \dot{x} \quad y \quad \dot{y} \quad z \quad \dot{z} \quad \phi \quad \dot{\phi} \quad \theta \quad \dot{\theta} \quad \psi \quad \dot{\psi}]^T$$

where u is the input vector and is defined as:

$$\boldsymbol{u} = [U_1 \quad U_2 \quad U_3 \quad U_4]^T$$

$$\ddot{x} = \frac{F}{m}\theta$$

$$\ddot{y} = \frac{F}{m}\phi$$

$$\ddot{z} = \frac{F}{m} - g$$

$$\ddot{\phi} = \frac{T_x}{I_x}$$

$$\ddot{\theta} = \frac{T_y}{I_y}$$

$$\ddot{\psi} = \frac{T_z}{I_z}$$

MODELLING OF SYSYEM

State-Space Representation

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{y} \\ \ddot{y} \\ \dot{z} \\ \ddot{z} \\ \dot{\phi} \\ \ddot{\phi} \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{\psi} \\ \ddot{\psi} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & g & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -g & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \\ \phi \\ \dot{\phi} \\ \theta \\ \dot{\theta} \\ \psi \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{I_x} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & 0 & \frac{1}{I_z} \end{bmatrix} \begin{bmatrix} F \\ T_x \\ T_y \\ T_z \end{bmatrix}$$

MODELLING OF SYSYEM

State-Space Representation

$$\begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \\ \phi \\ \dot{\phi} \\ \theta \\ \dot{\theta} \\ \psi \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \\ \phi \\ \dot{\phi} \\ \theta \\ \dot{\theta} \\ \psi \\ \dot{\psi} \end{bmatrix}$$

MODELLING OF SYSYEM

BLDC Motor System Identification

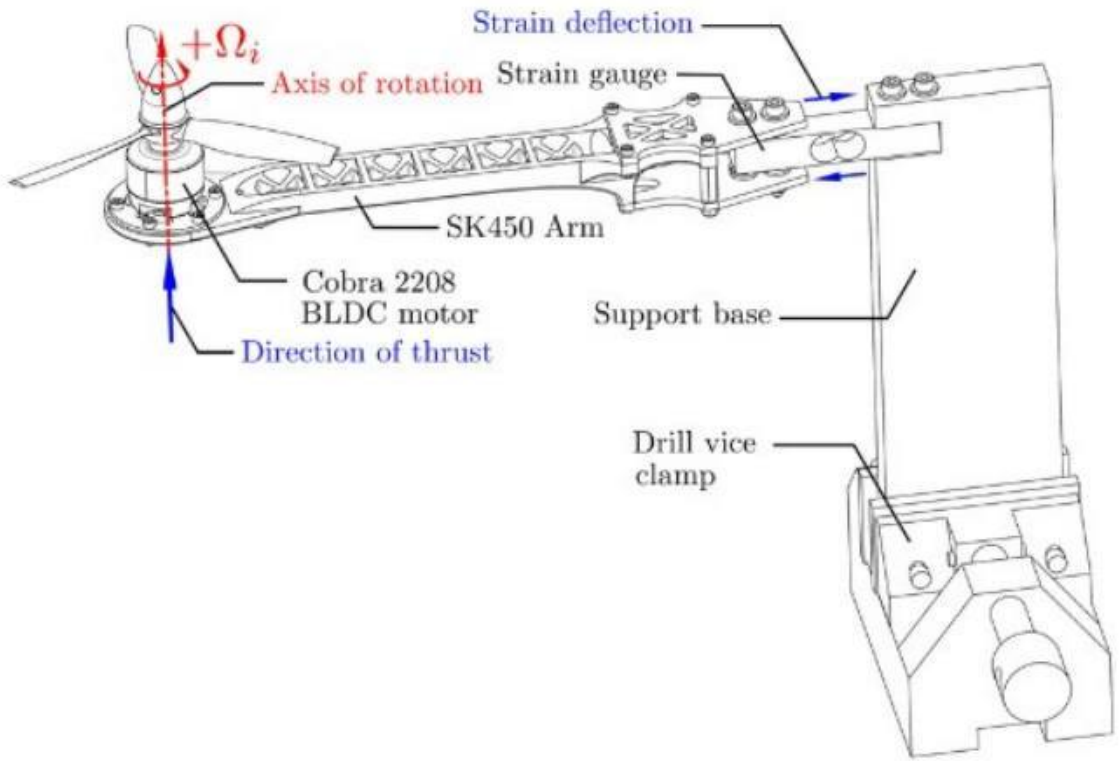


Figure 1:Diagram of the setup created for thrust calculation of a propeller-driven BLDC motor.

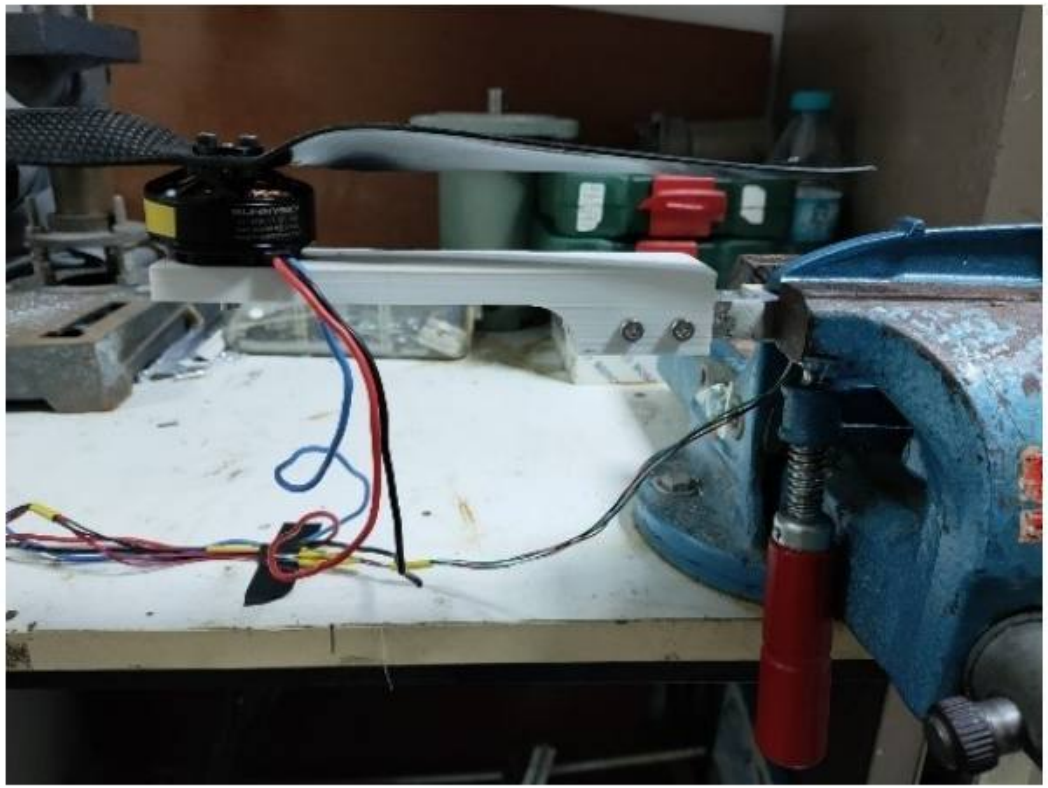


Figure 1:The setup designed for thrust calculation of a propeller-driven BLDC motor.

MODELLING OF SYSYEM

BLDC Motor System Identification

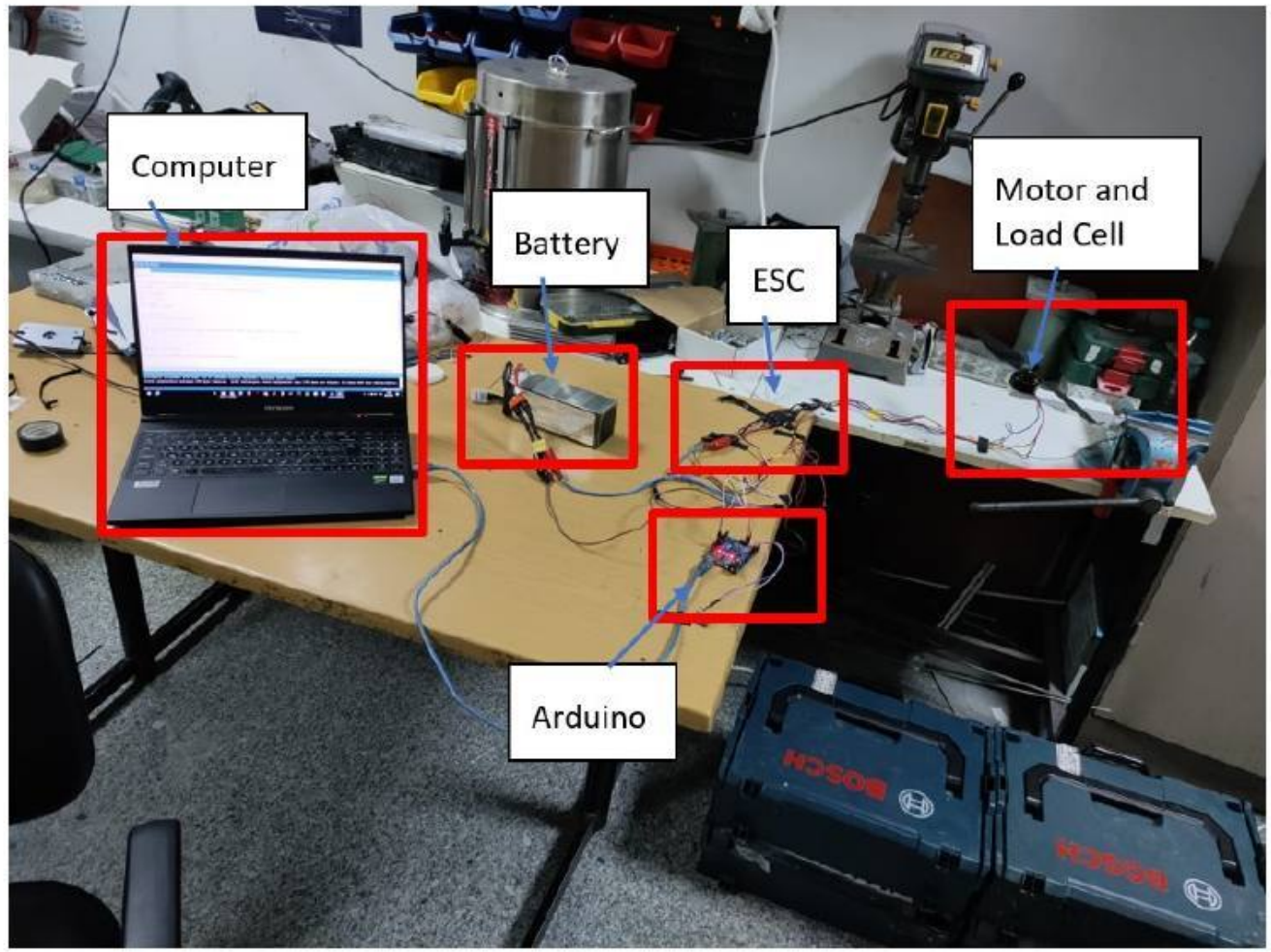
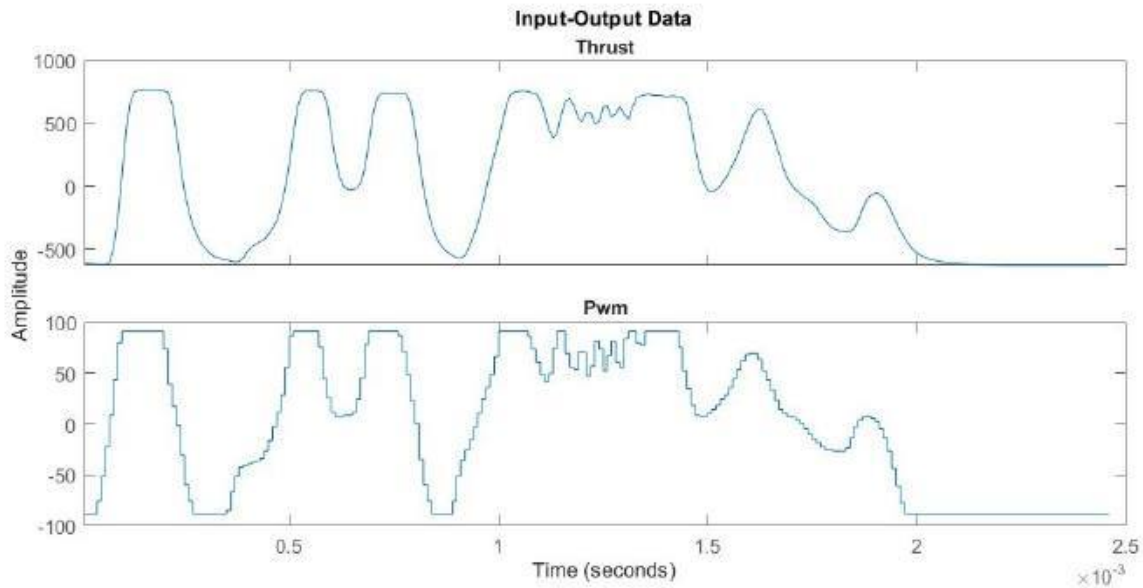


Figure 1:The components of the setup designed for thrust calculation of a propeller-driven BLDC motor

MODELLING OF SYSYEM

BLDC Motor System Identification



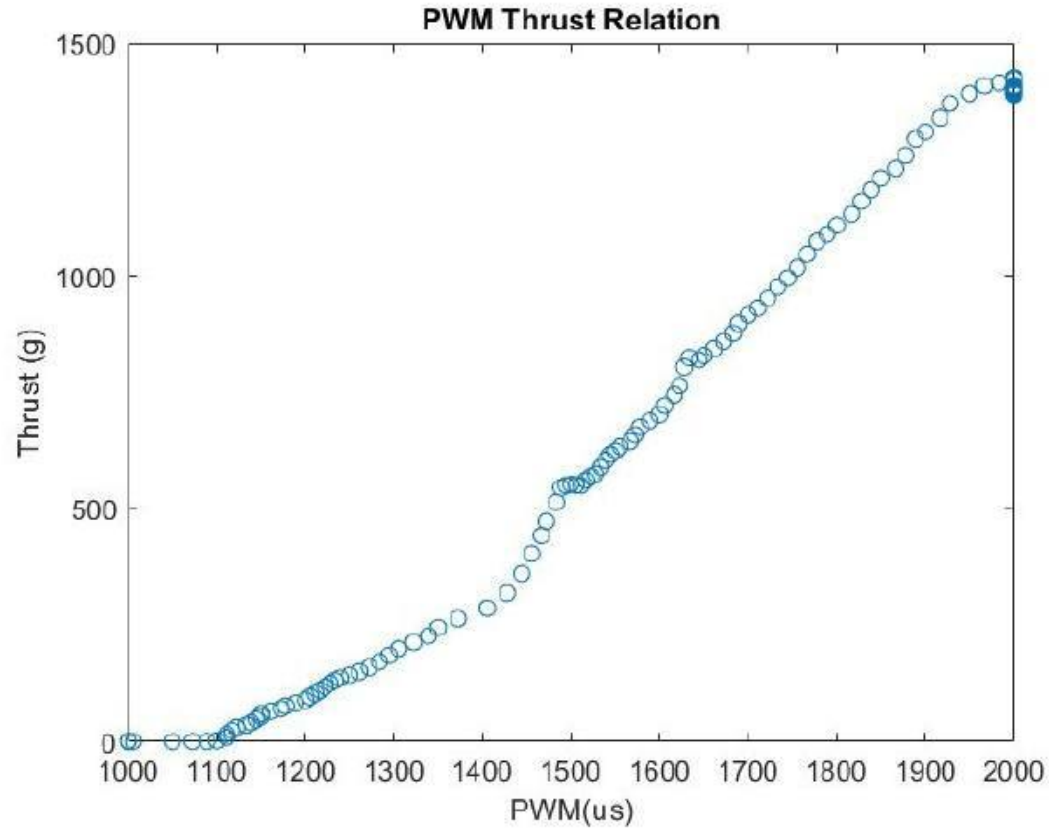
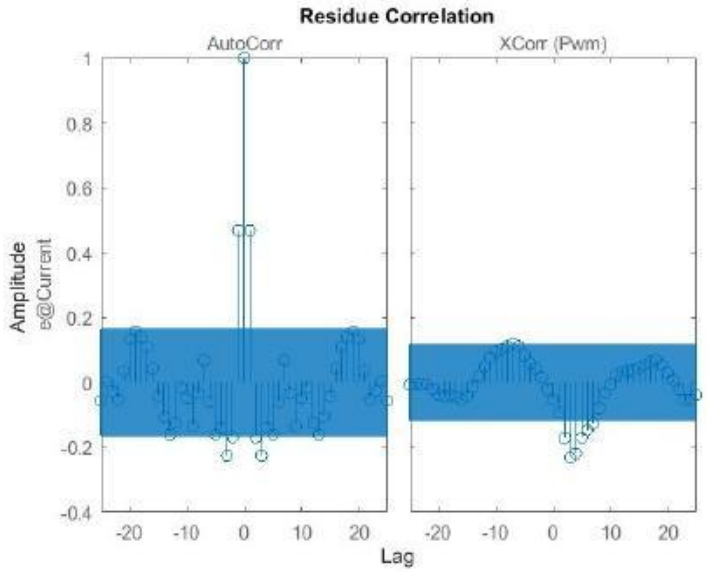
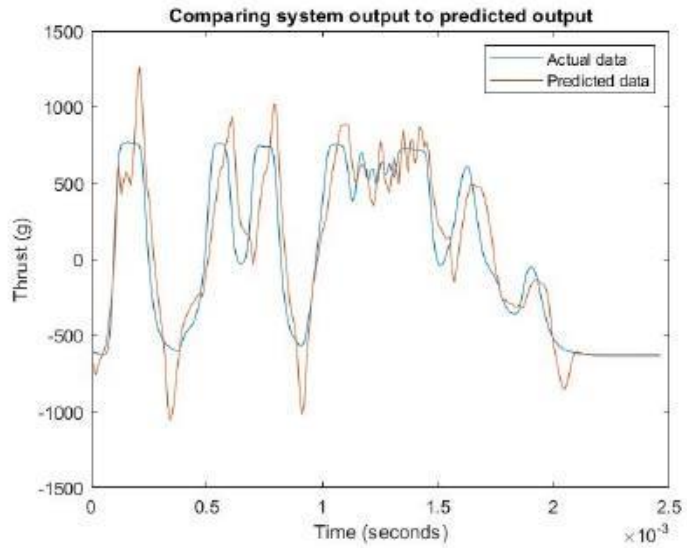


Figure 1:The graph of the dataset obtained from input PWM and output thrust

MODELLING OF SYSYEM

BLDC Motor System Identification





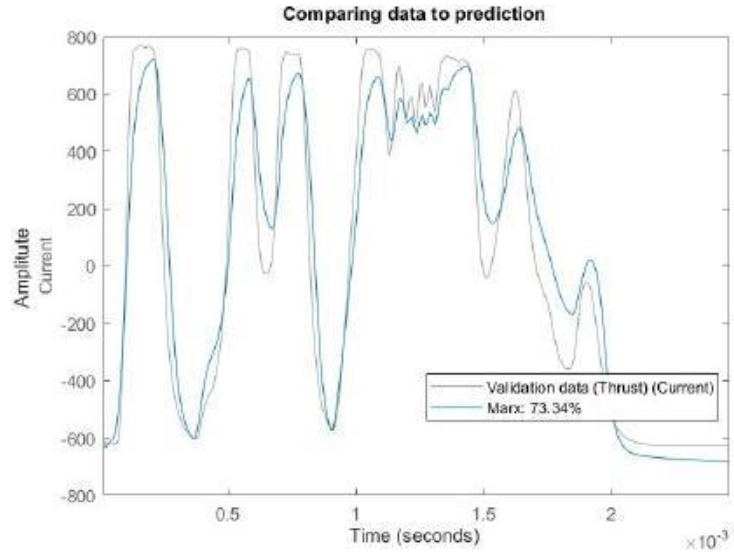


Figure 1:System identification output graphs

MODELLING OF SYSYEM

BLDC Motor System Identification

Command Window

```
Marx =
Discrete-time ARX model: A(z)y(t) = B(z)u(t) + e(t)

  A(z) = 1 - 1.767 (+/- 0.06345) z^-1 + 0.8461 (+/- 0.1176) z^-2
                                      - 0.07109 (+/- 0.06225) z^-3

  B(z) = 2.073 (+/- 0.3494) z^-2 - 2.013 (+/- 0.3531) z^-3

Sample time: 1e-05 seconds

Parameterization:
   Polynomial orders:   na=3   nb=2   nk=2
   Number of free coefficients: 5
   Use "polydata", "getpvec", "getcov" for parameters and their uncertainties.

Status:
Estimated using ARX on time domain data "ze".
Fit to estimation data: 98.74% (prediction focus)
FPE: 42.15, MSE: 39.49
More information in model's "Report" property.

Model Properties

marxPoles =

    0.9727
    0.6886
    0.1061
```

```
>> transfer_function

transfer_function =

            2.073 z - 2.013
  -----------------------------------
  z^3 - 1.767 z^2 + 0.8461 z - 0.07109

Sample time: 1e-05 seconds
Discrete-time transfer function.
```

Figure 1:The discrete-time transfer function obtained for the BLDC motor

MODELLING OF SYSYEM

BLDC Motor System Identification

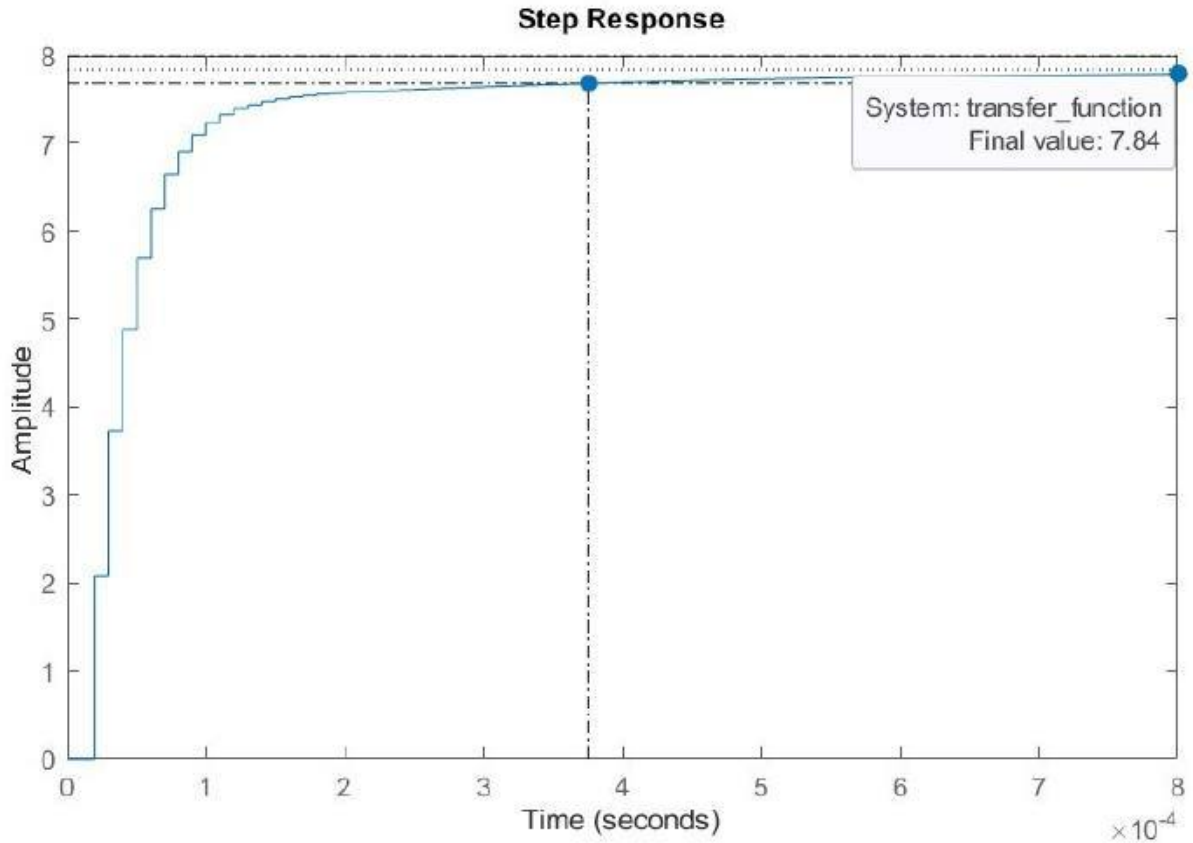


Figure 1: The output obtained when a step input is applied to the transfer function of the BLDC motor

DESIGN AND BUILD

Mechanical Design

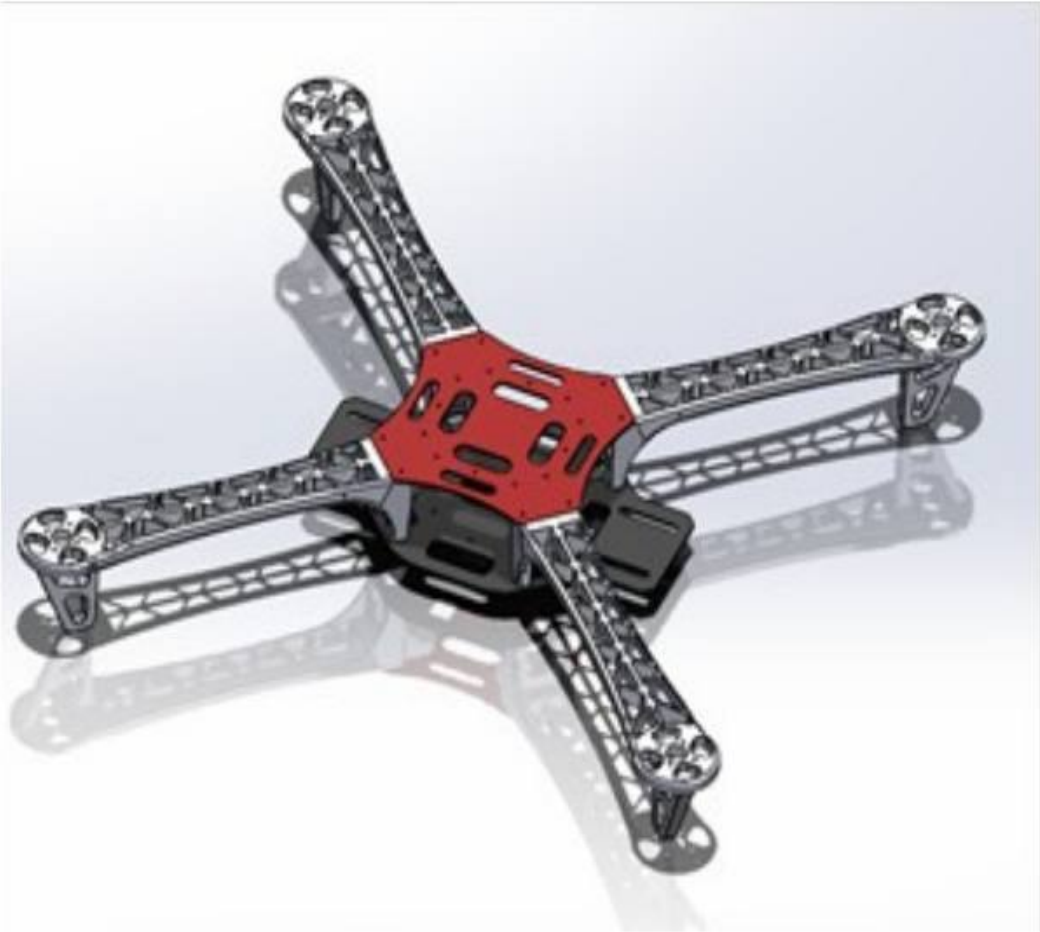




Figure 1: CAD design and mechanical design images of the drawn chassis

DESIGN AND BUILD

Hardware Component and Integration

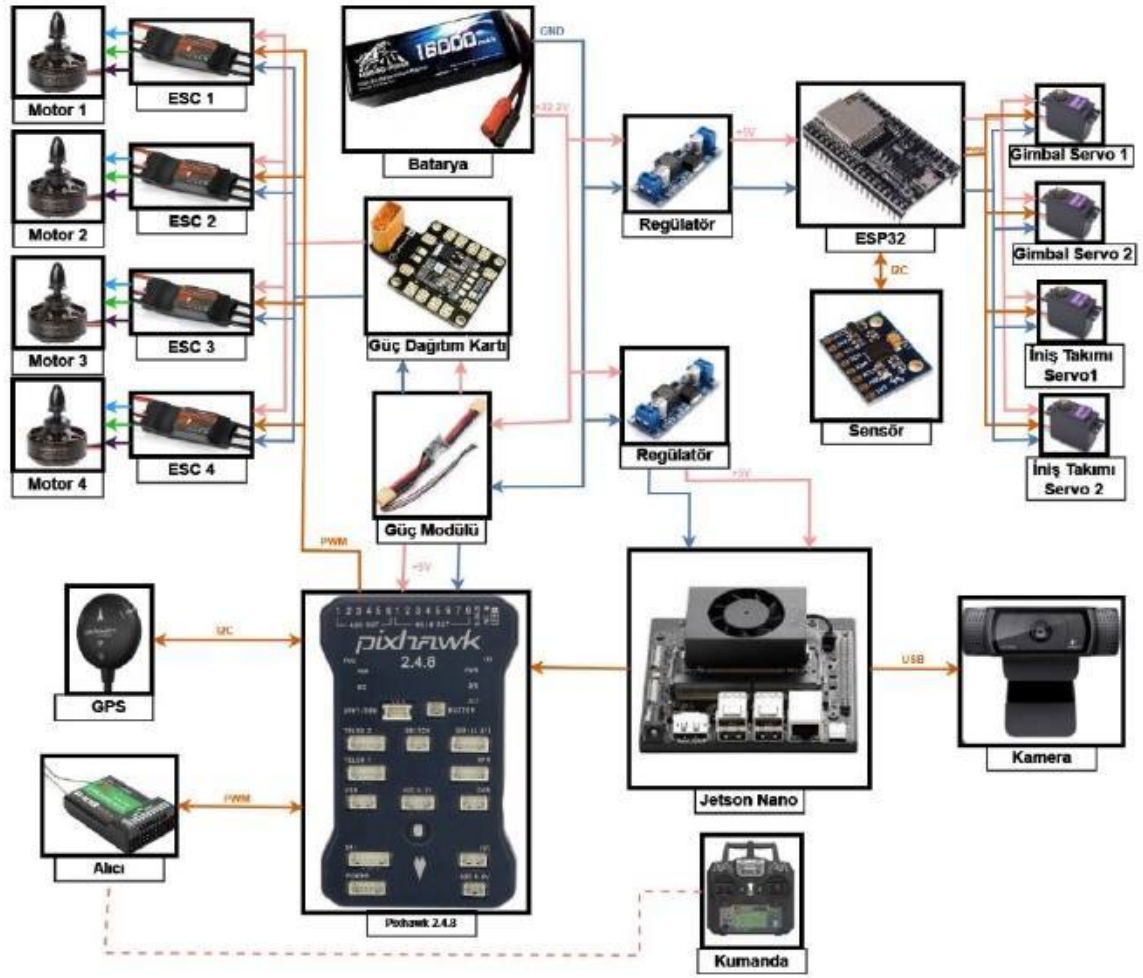


Figure 1: Hardware diagram of the quadrotor

CONTROL METHOD

P-PID CASCADE CONTROL

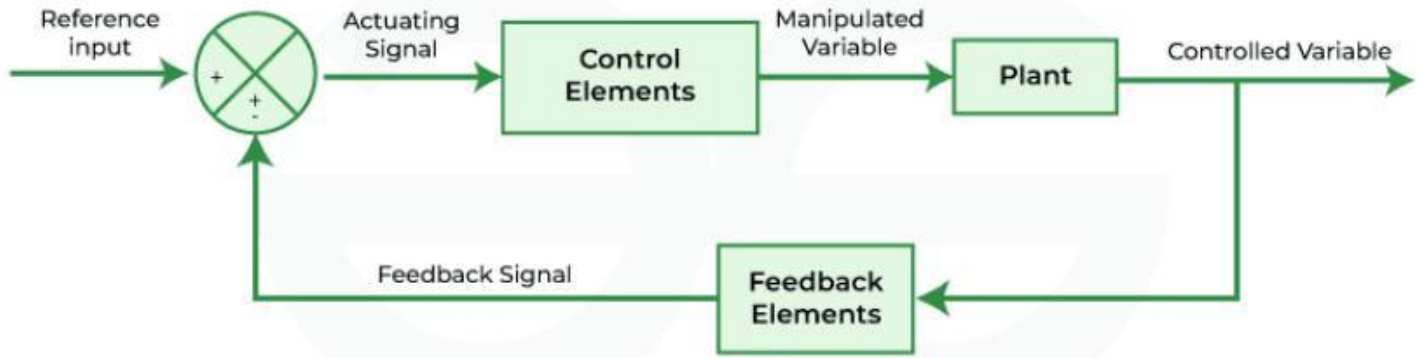


Figure 4: General Control System Block Diagram

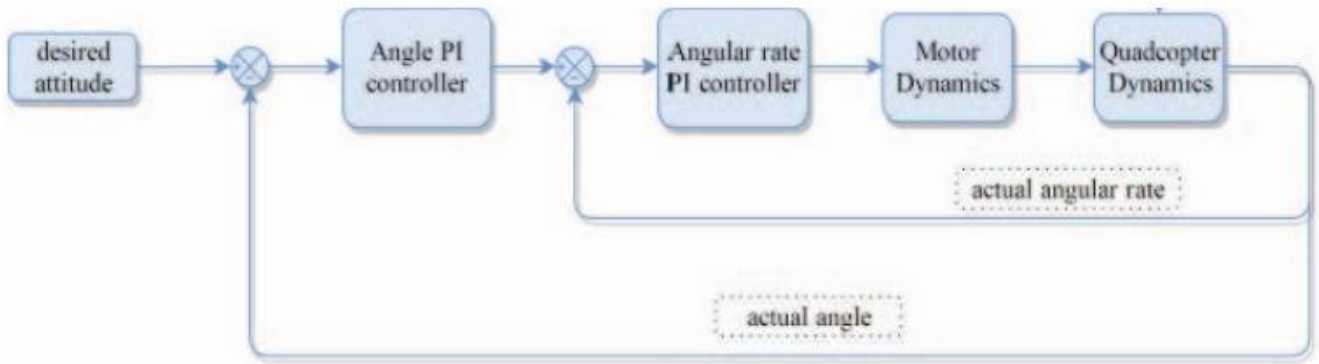


Figure 4: Cascade Attitute Control System Block Diagram

CONTROL METHOD

P-PID CASCADE CONTROL

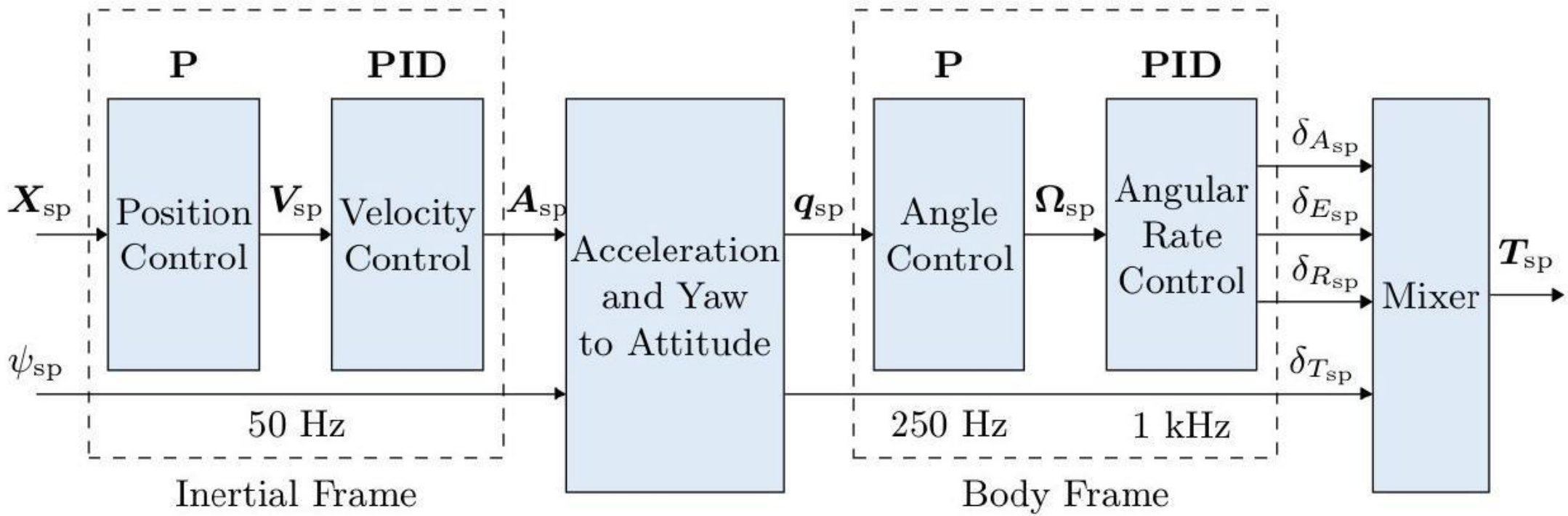


Figure 4: Cascade Attitute and Position Control System Block Diagram

CONTROL METHOD

Controller Design with Simulink

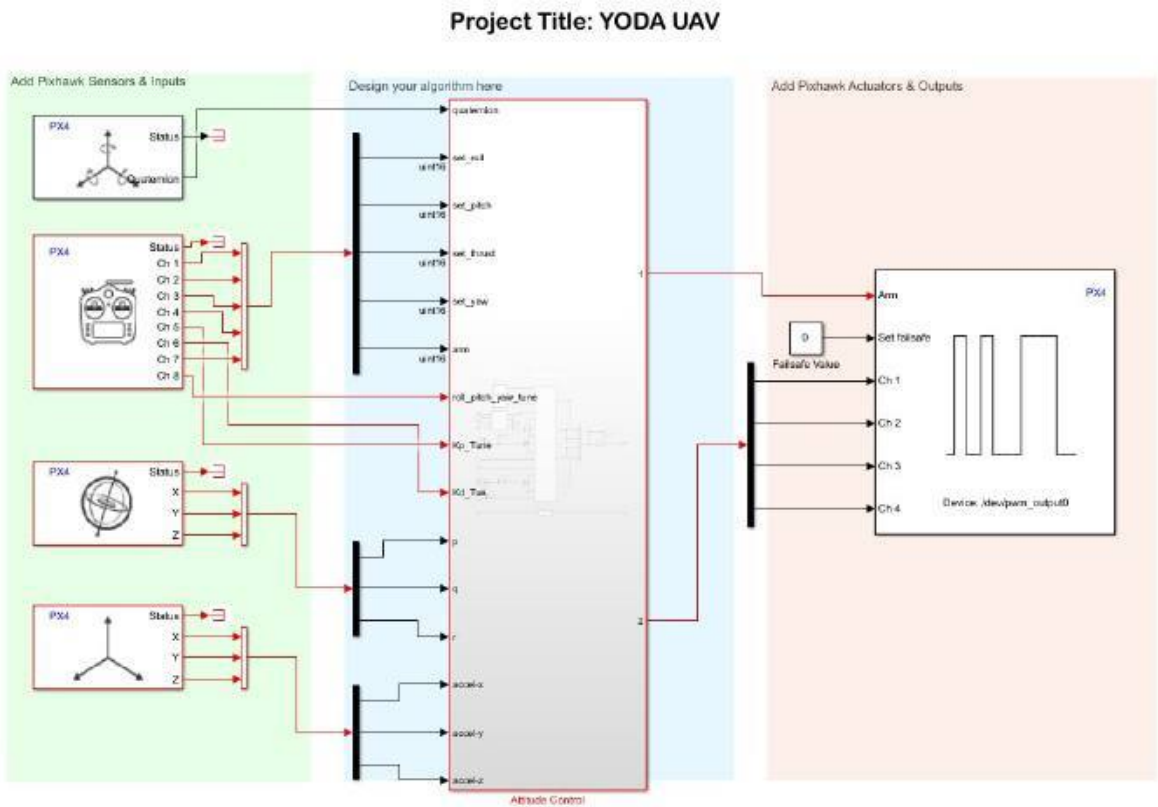


Figure 4: Overall System Simulink Diagram

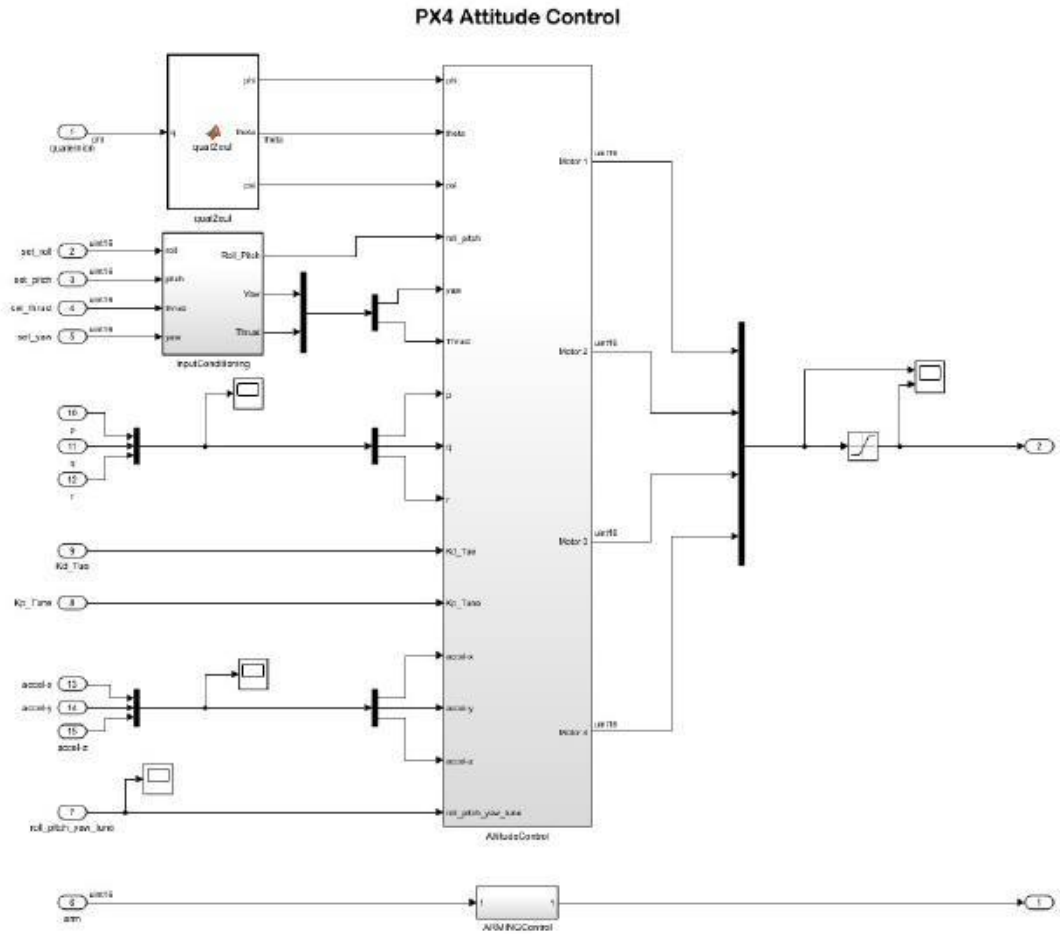


Figure 4: Attitute Controller Subsystem Simulink Diagram

CONTROL METHOD

Controller Design with Simulink

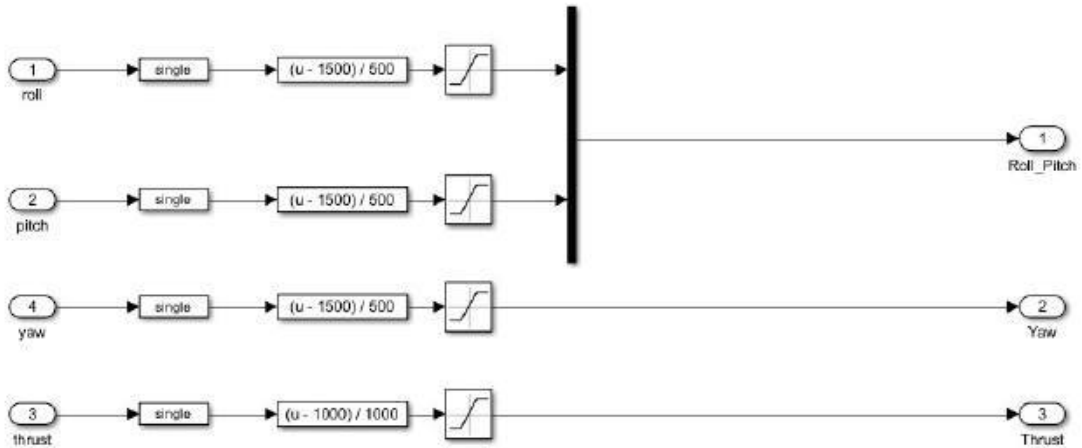


Figure 4: Input Condition Subsystem Simulink Diagram



Figure 4: Radio Controller Simulink and QGroundControl Interface

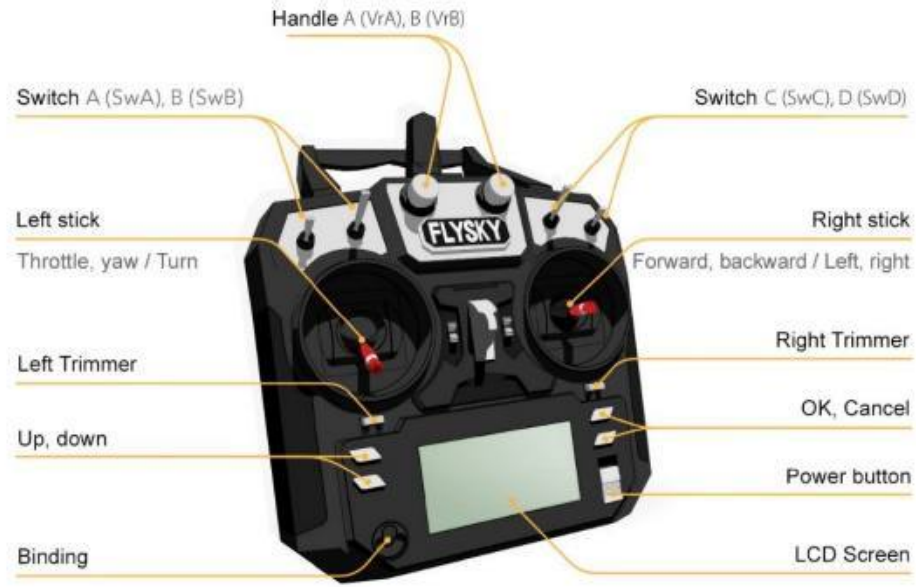


Figure 4: Radio Controller Image

CONTROL METHOD

Controller Design with Simulink

quat2eul

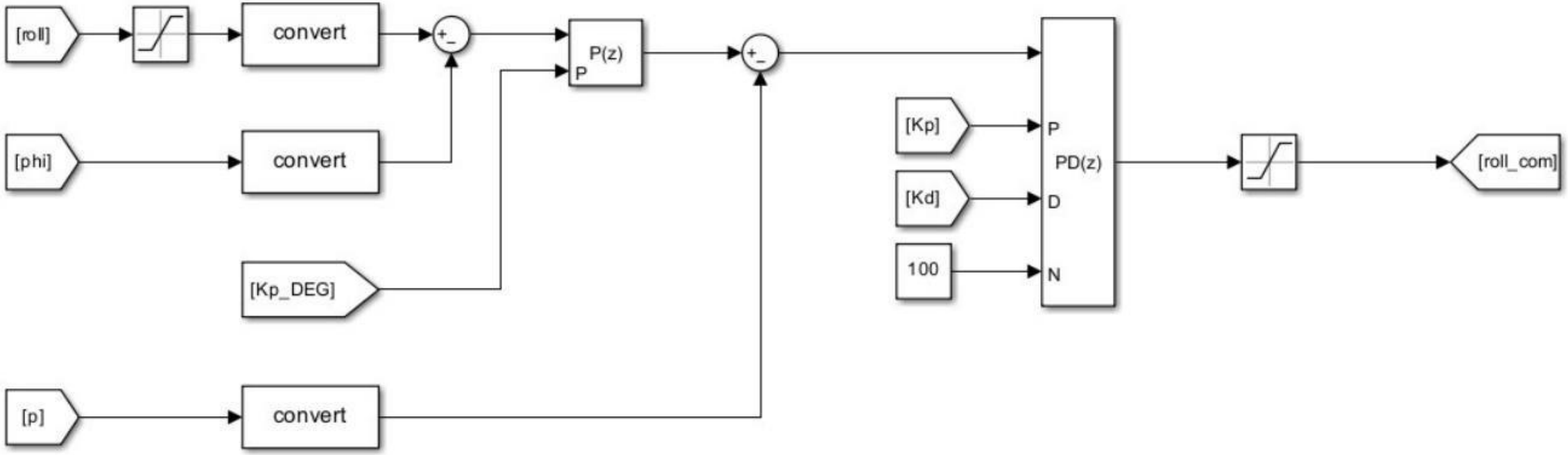
```
1   function [phi,theta,psi] = quat2eul(q)
2
3   % Quaternion conversion to Euler angles based on the PX4 Firmware v1.6.5
4   % release.
5
6   phi = atan2(2*(q(1)*q(2) + q(3)* q(4)), q(1)^2 - q(2)^2 - q(3)^2 + q(4)^2);
7   theta = asin(-2*(q(2)*q(4) - q(1)*q(3)));
8   psi = atan2(2*(q(2)*q(3) + q(1)* q(4)), q(1)^2 + q(2)^2 - q(3)^2 - q(4)^2);
9
10  end
```

Figure 4: Quaternion to Euler Matlab Function Block

$$\begin{bmatrix}\varphi\\ \theta\\ \psi\end{bmatrix} = \begin{bmatrix} atan2\left(2,\left(q_w q_x + q_y q_z\right)1 - 2\left(q_x^2 + q_y^2\right)\right) \\ -\frac{\pi}{2} + 2atan2\left(\sqrt{1 + 2\left(q_w q_y - q_x q_z\right)}, \sqrt{1 - 2\left(q_w q_y - q_x q_z\right)}\right) \\ atan2\left(2,\left(q_w q_z + q_x q_y\right)1 - 2\left(q_y^2 + q_z^2\right)\right) \end{bmatrix}$$

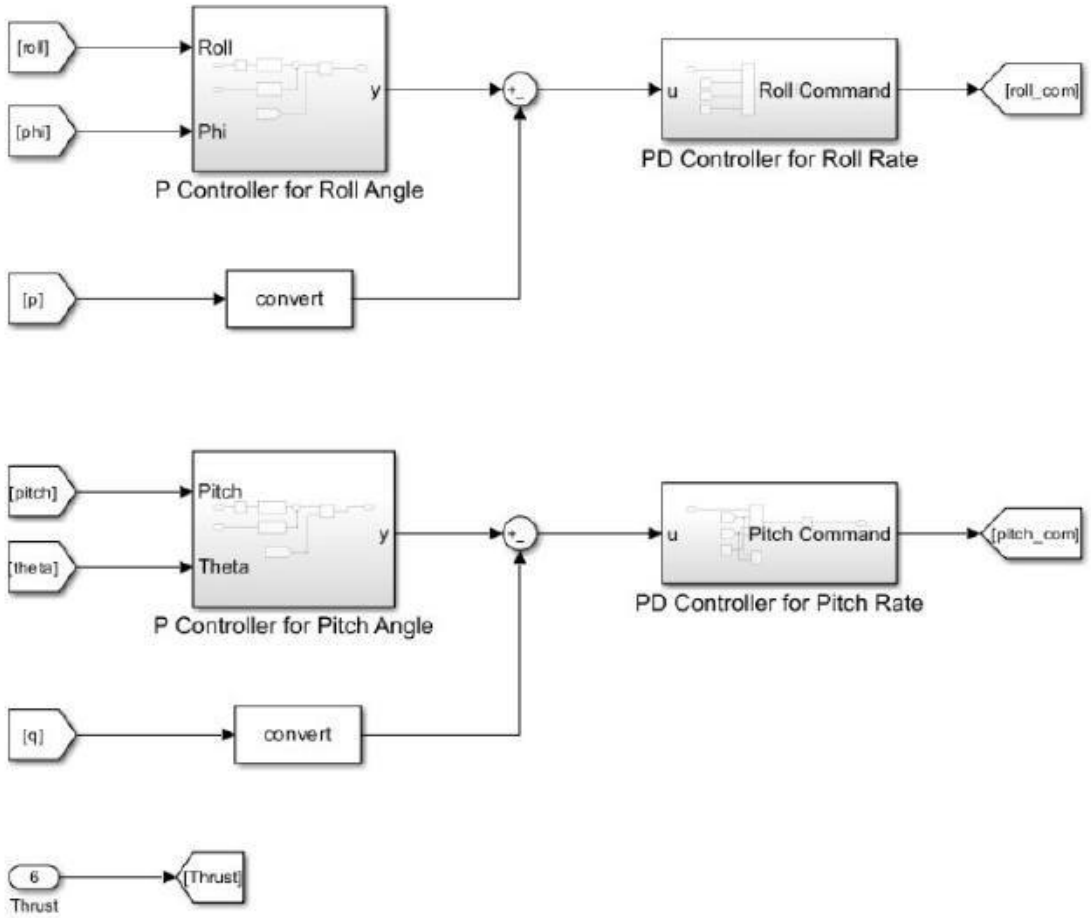
CONTROL METHOD

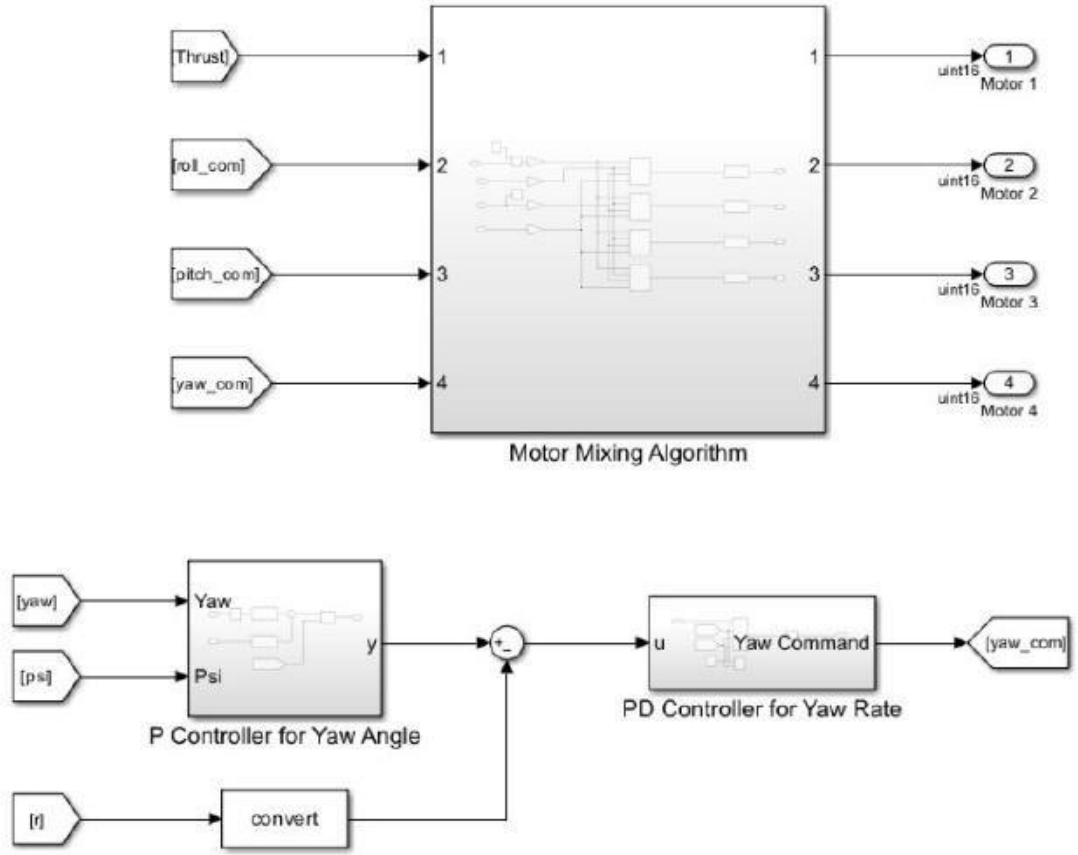
Controller Design with Simulink



CONTROL METHOD

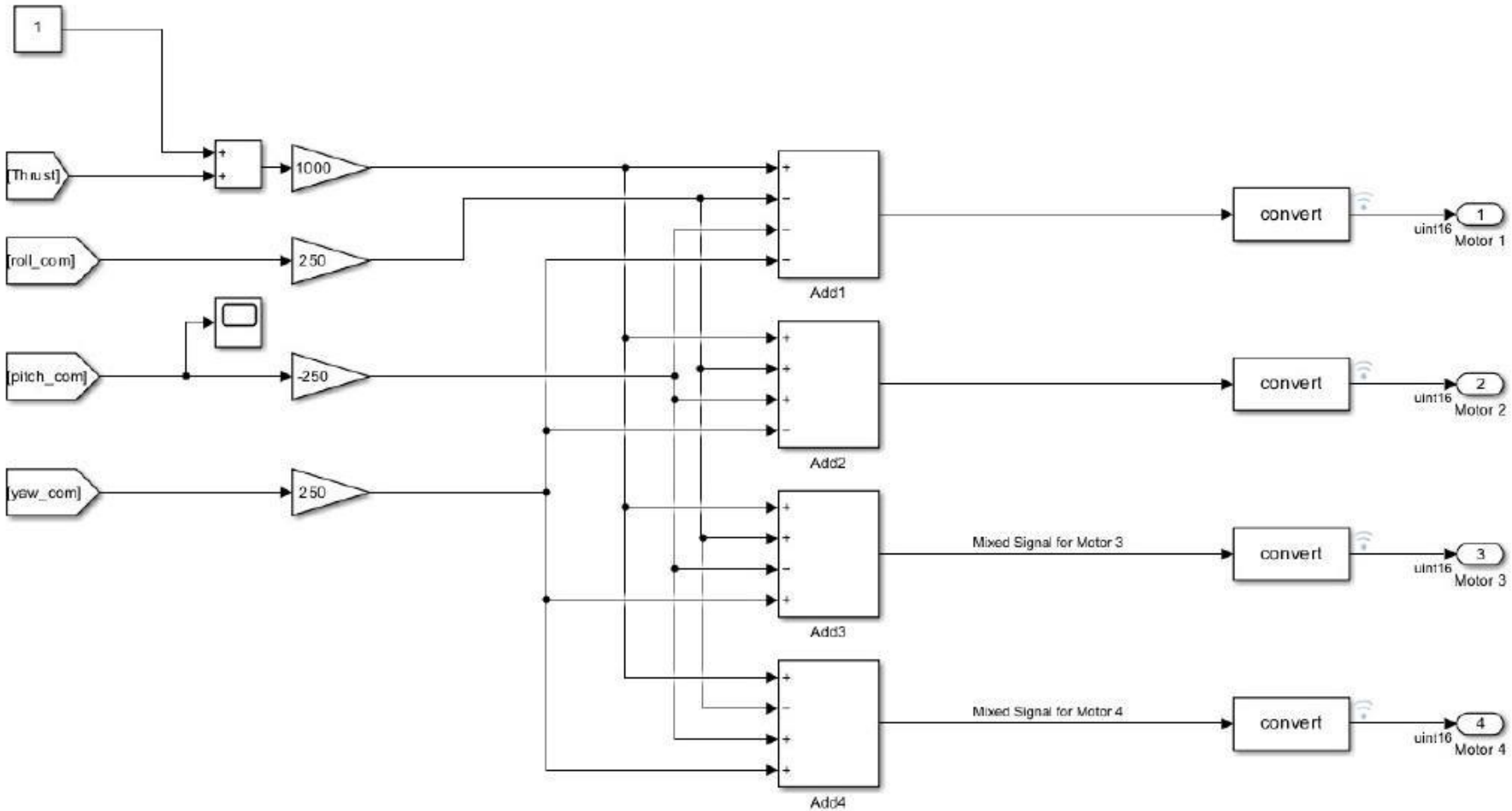
Controller Design with Simulink





CONTROL METHOD

Controller Design with Simulink

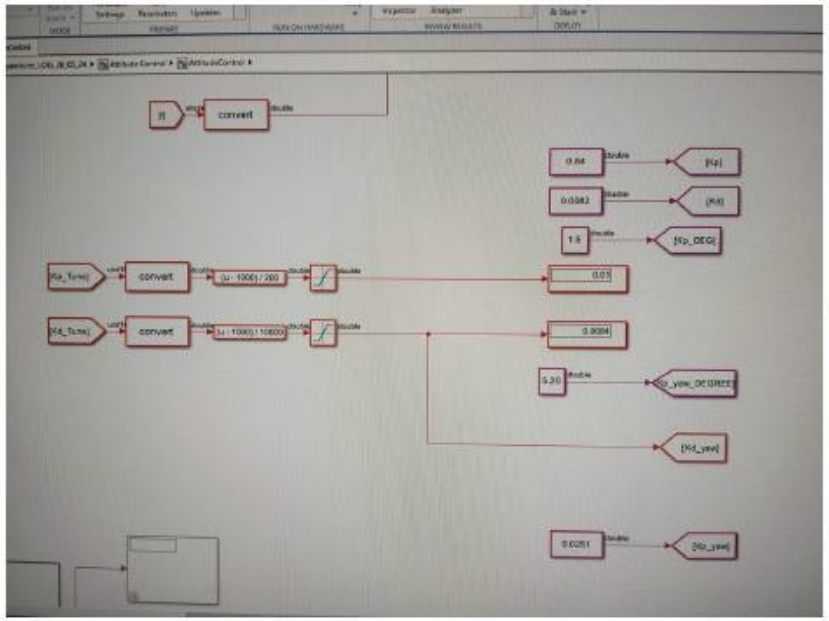


CONTROL METHOD

Tuning of the Controller Coefficient



Figure 4: Radio Controller Image



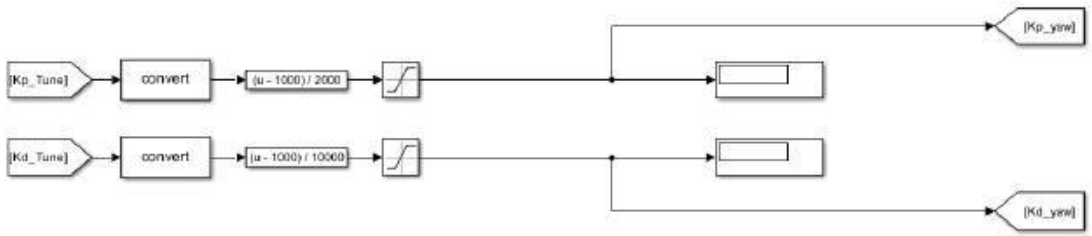
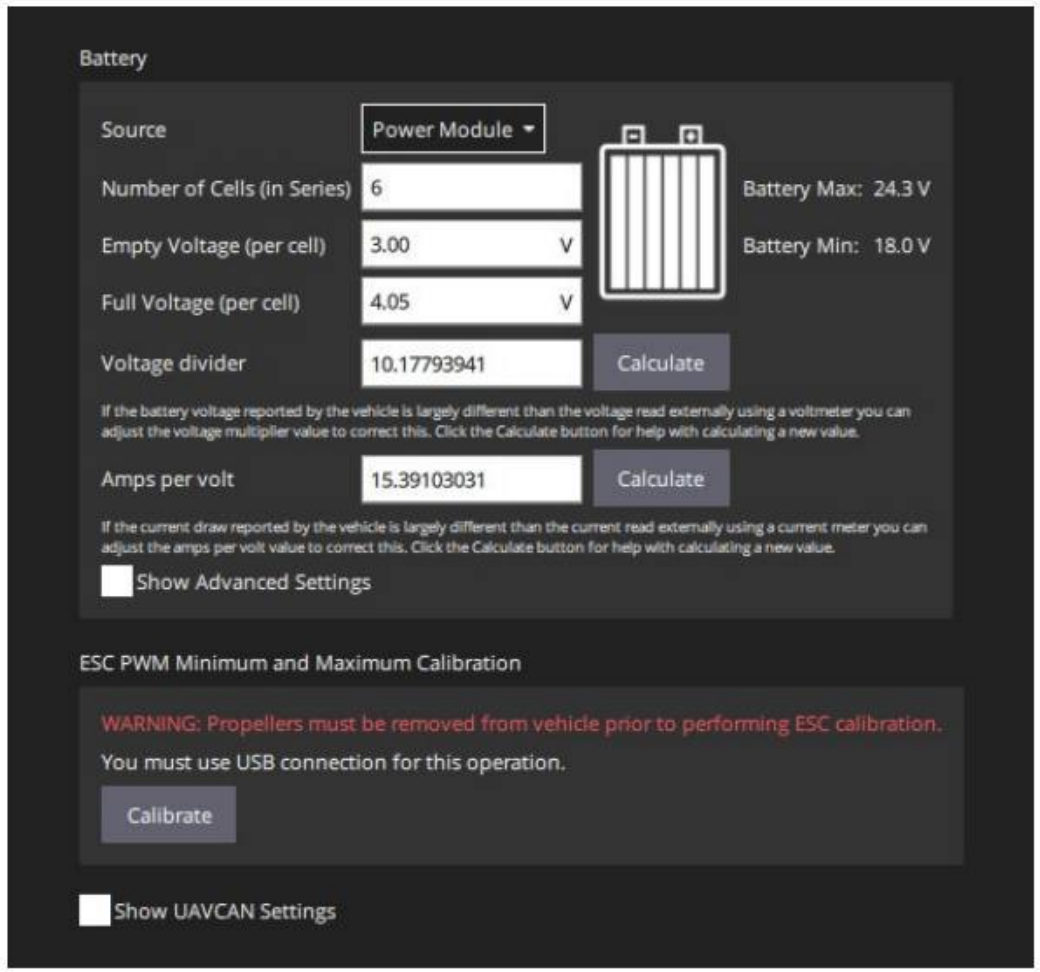


Figure 4: Tuning Simulink Diagram

IMPLEMENTATION OF HARDWARE

Pixhawk Calibrations



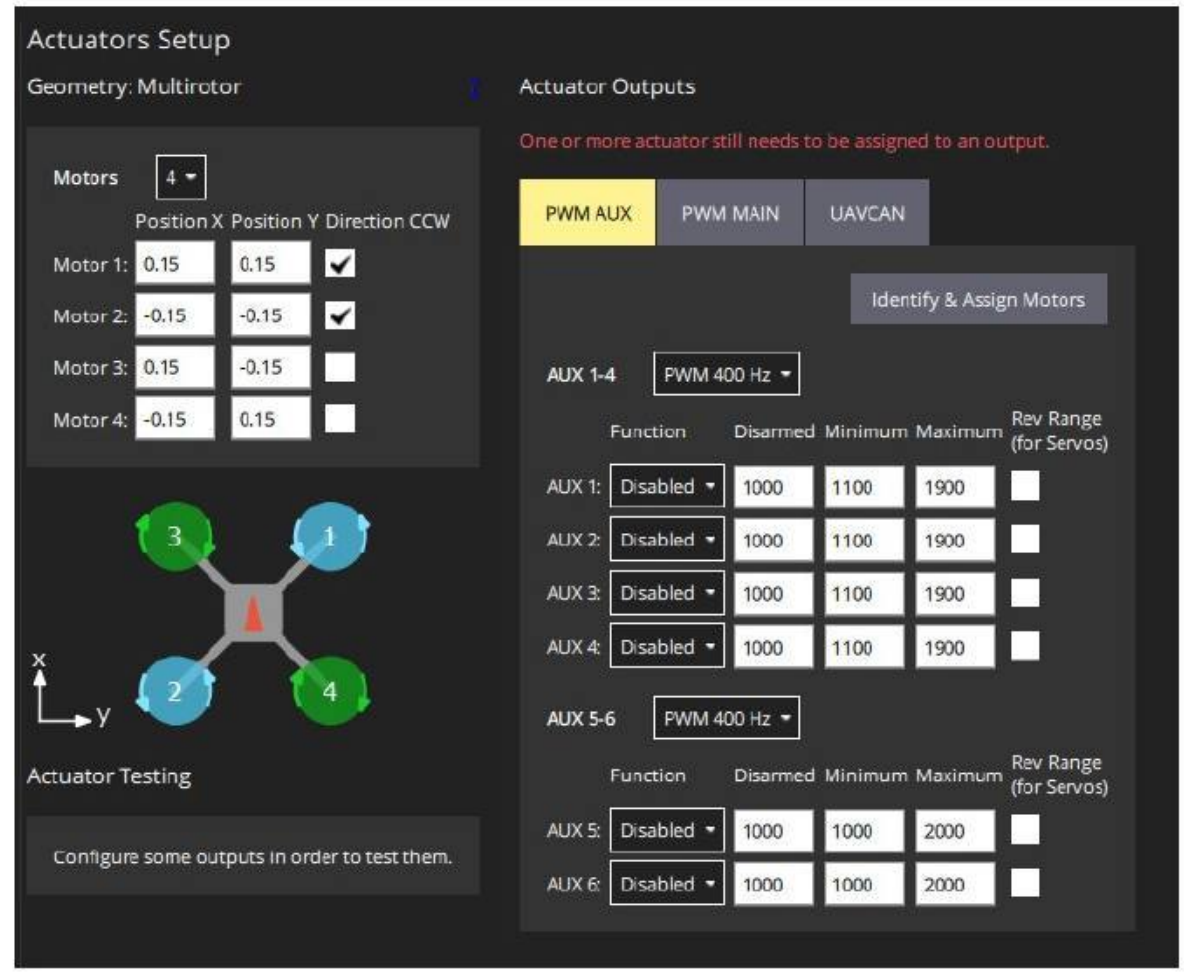
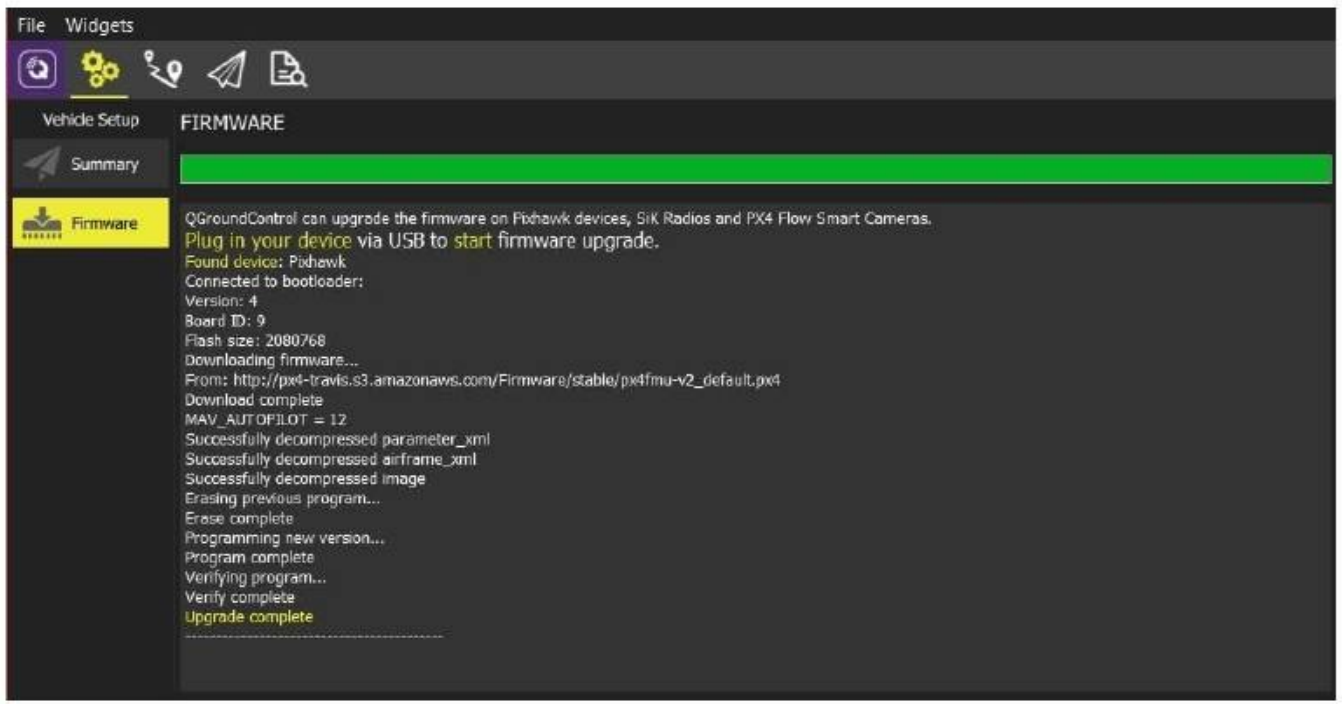


Figure 4: Calibration interfaces performed via QGround Control program

IMPLEMENTATION OF HARDWARE

Firmware Deploy



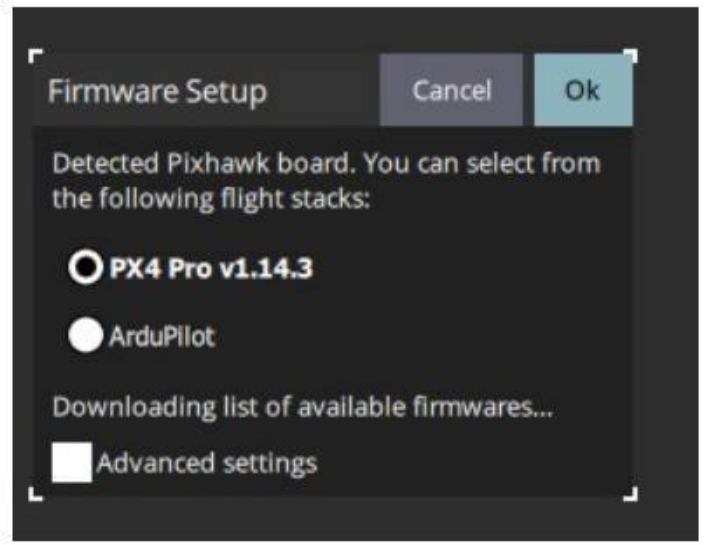


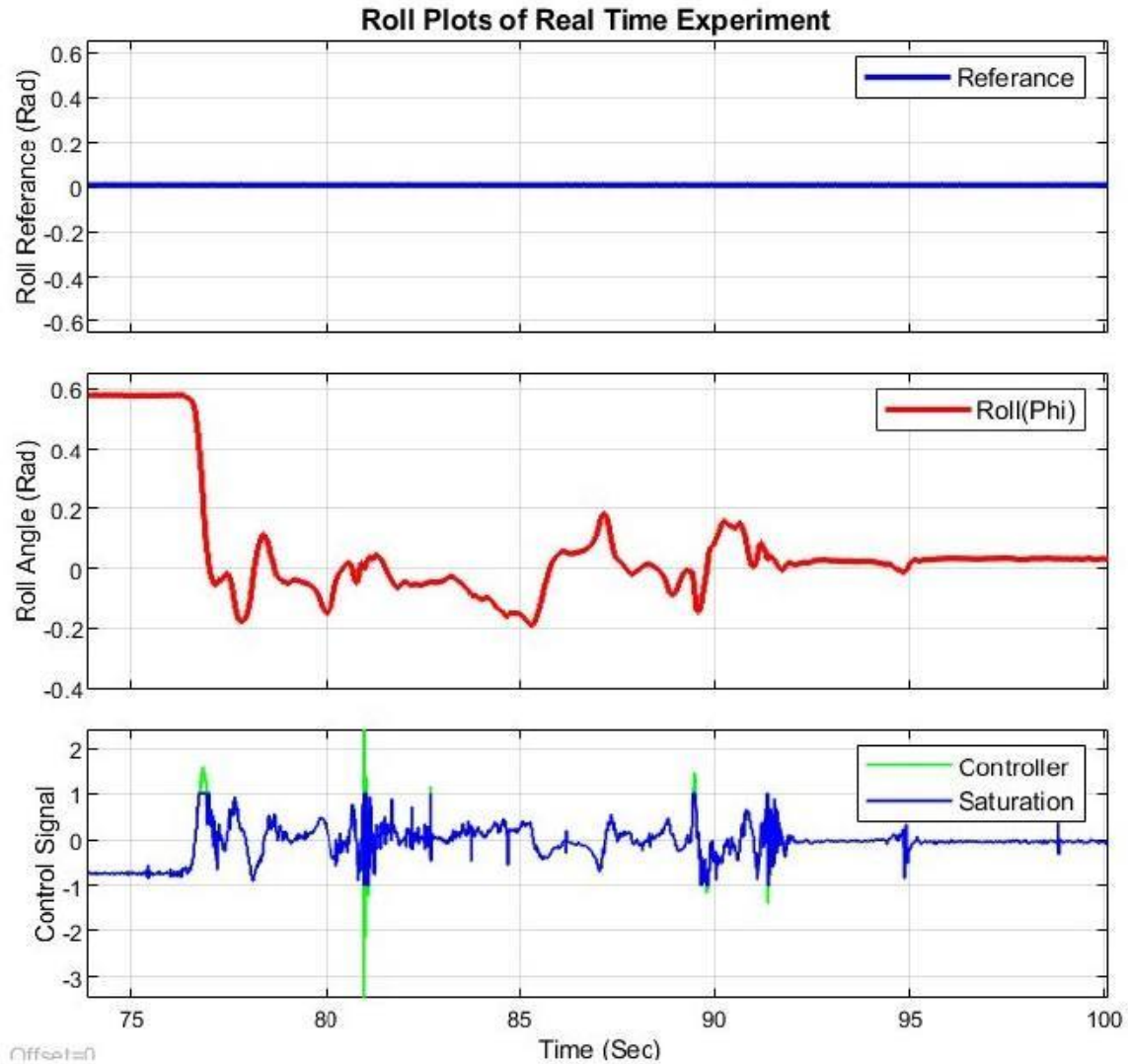


Figure 4: Software deployment interfaces on QGround and Matlab

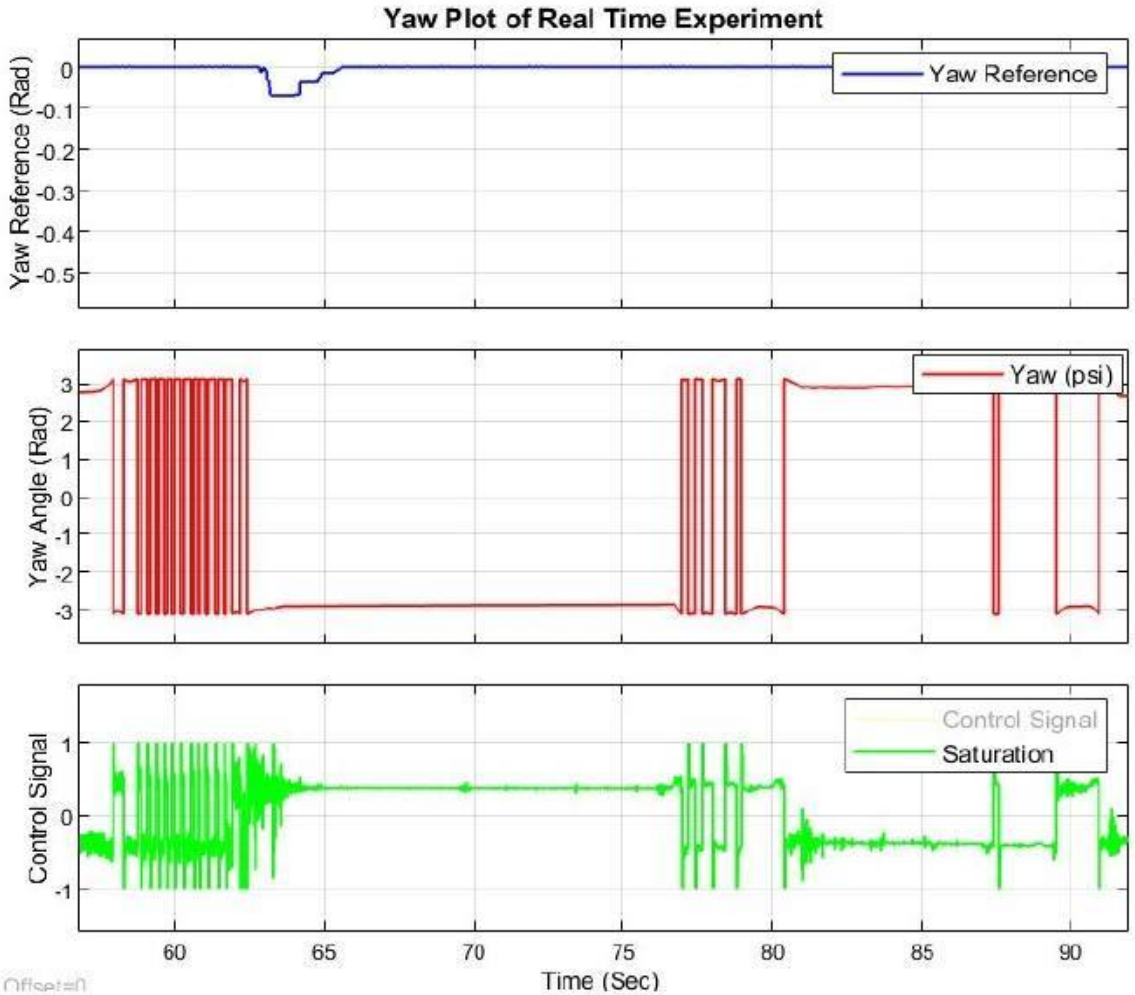
RESULTS AND DISCUSSIONS

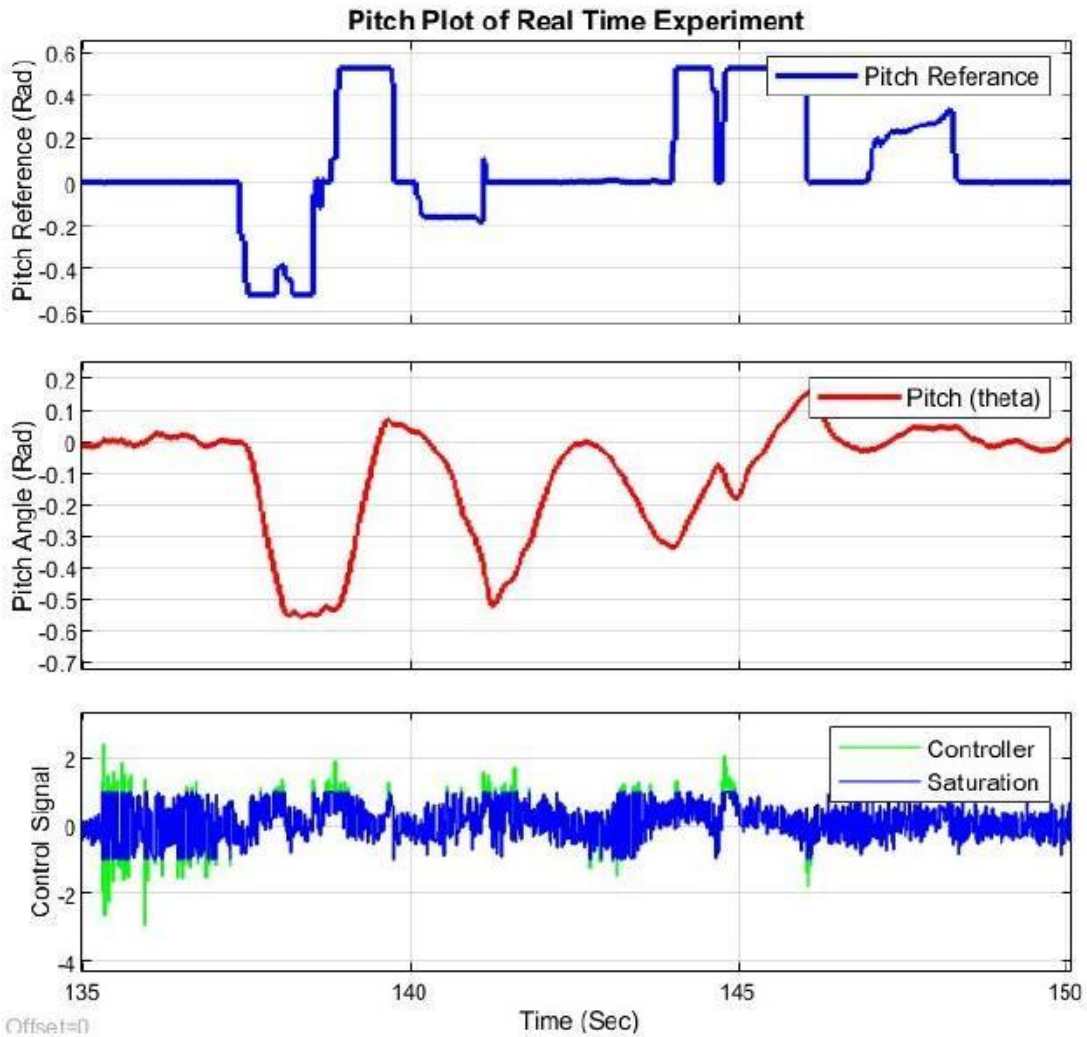


Figure 4:The initial position of the quadrotor

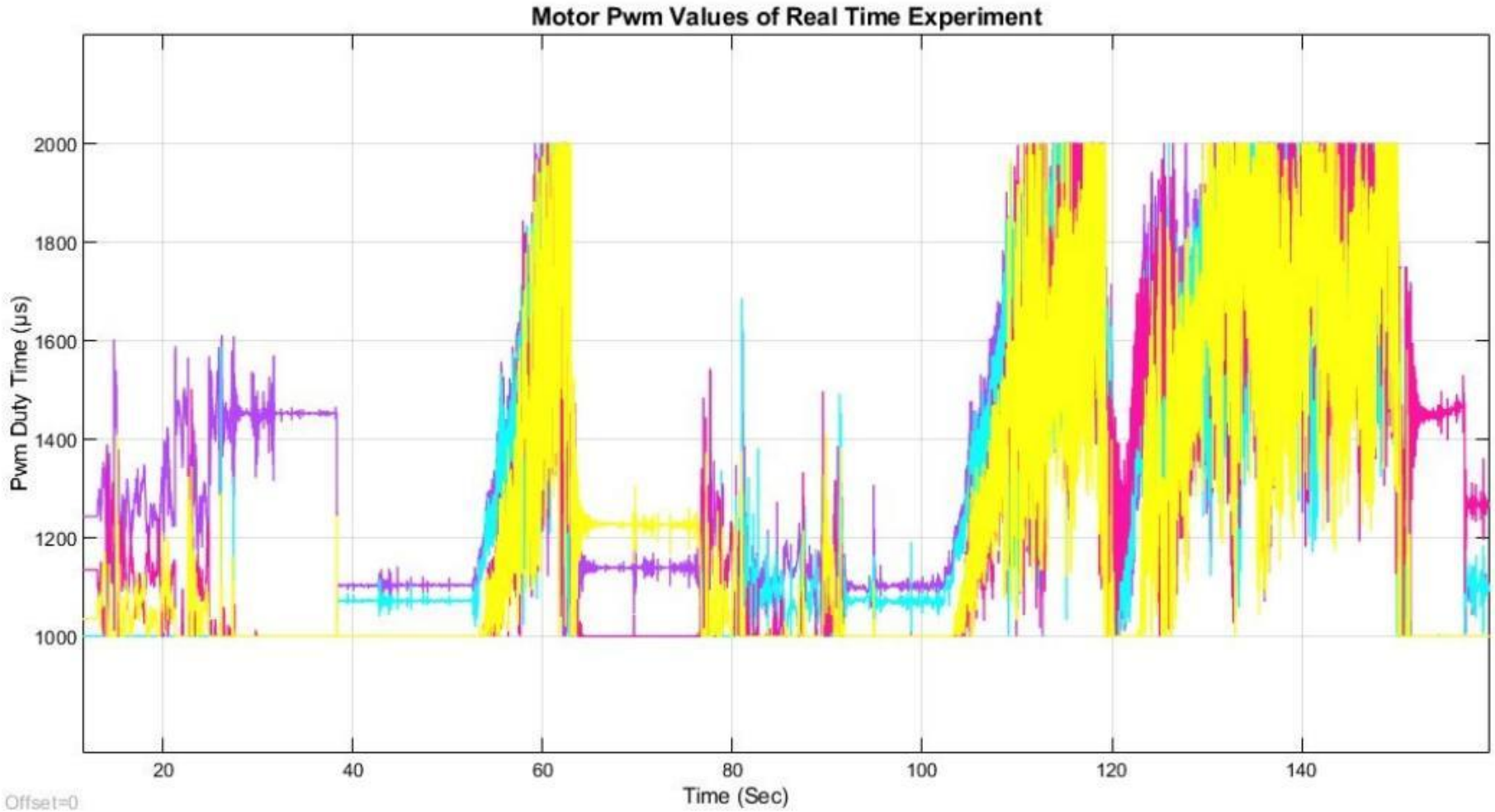


RESULTS AND DISCUSSIONS





RESULTS AND DISCUSSIONS



RESULTS AND DISCUSSIONS

System movements without a controller



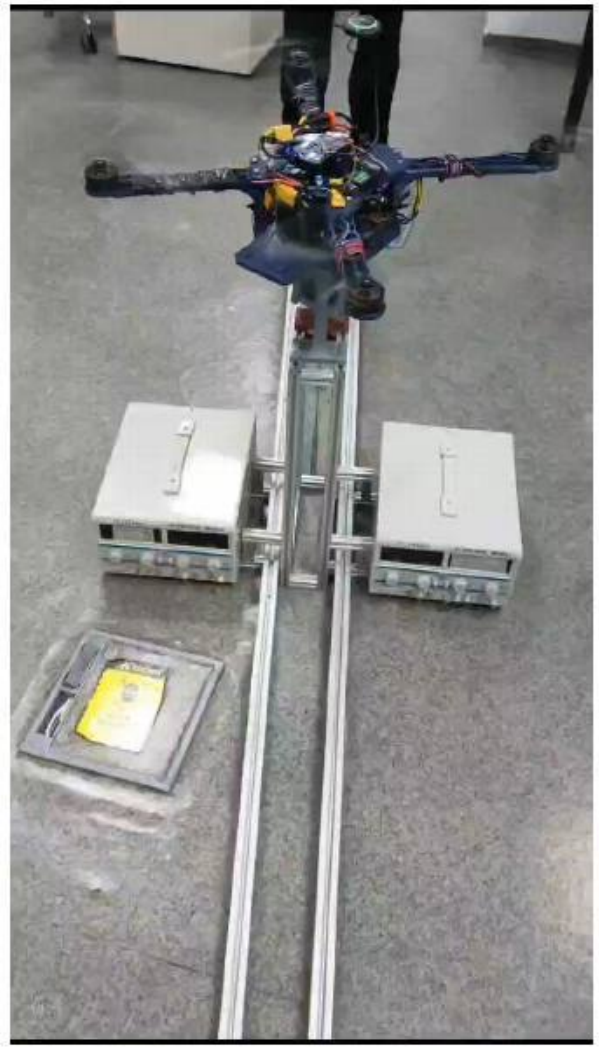
RESULTS AND DISCUSSIONS

System movements with only a PD controller.



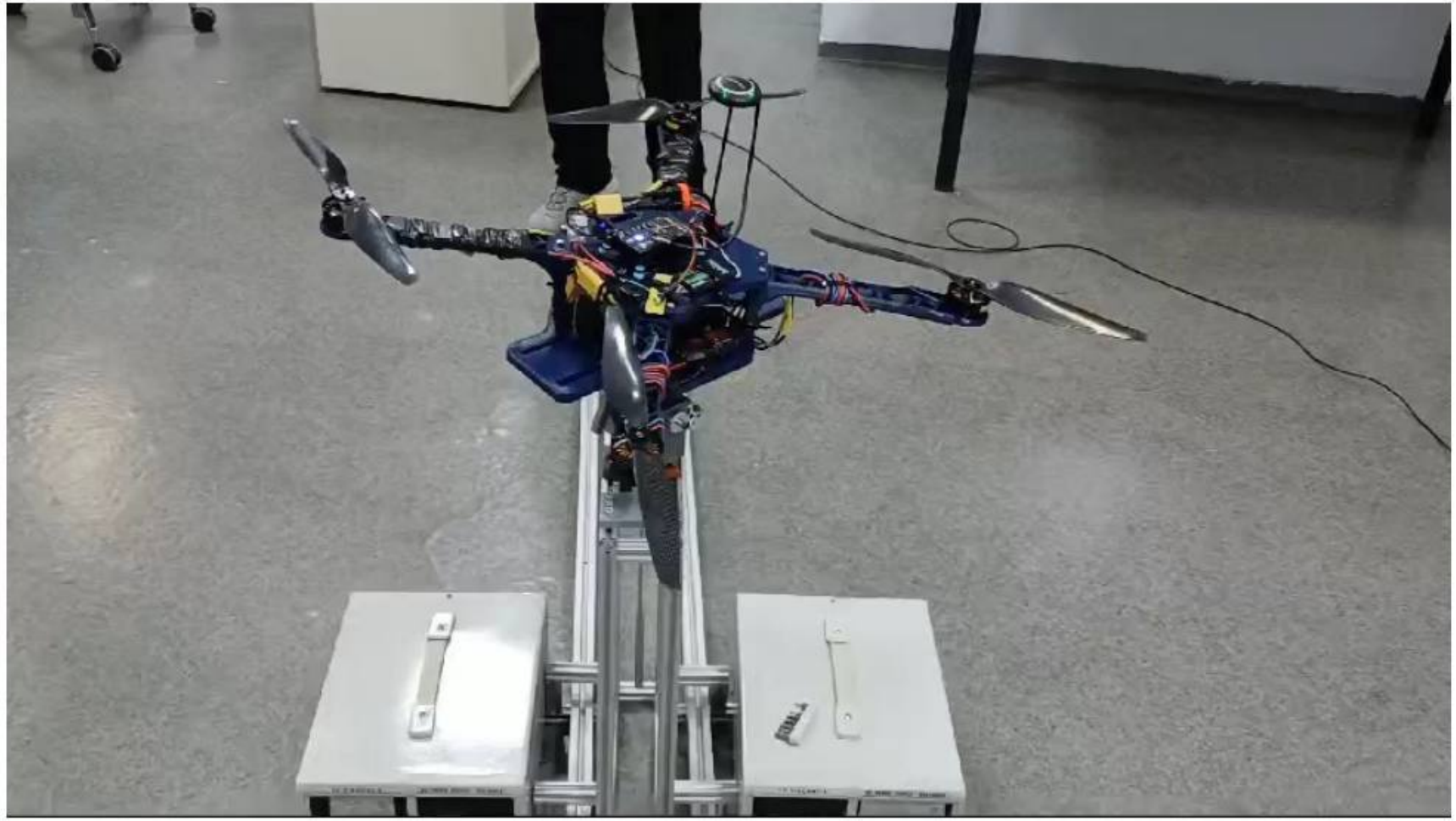
RESULTS AND DISCUSSIONS

System movement when only the yaw controller is absent.



RESULTS AND DISCUSSIONS

After designing the P-PD controller, system movements.



RESULTS AND DISCUSSIONS

Fail ☺



THE FUTURE OBJECTİVES OF THE PROJECT

SIMULATION



Figure 1:Image of logo detection using YOLO in the area where the rotary-wing unmanned aerial vehicle will land..

THE FUTURE OBJECTİVES OF THE PROJECT

SIMULATION

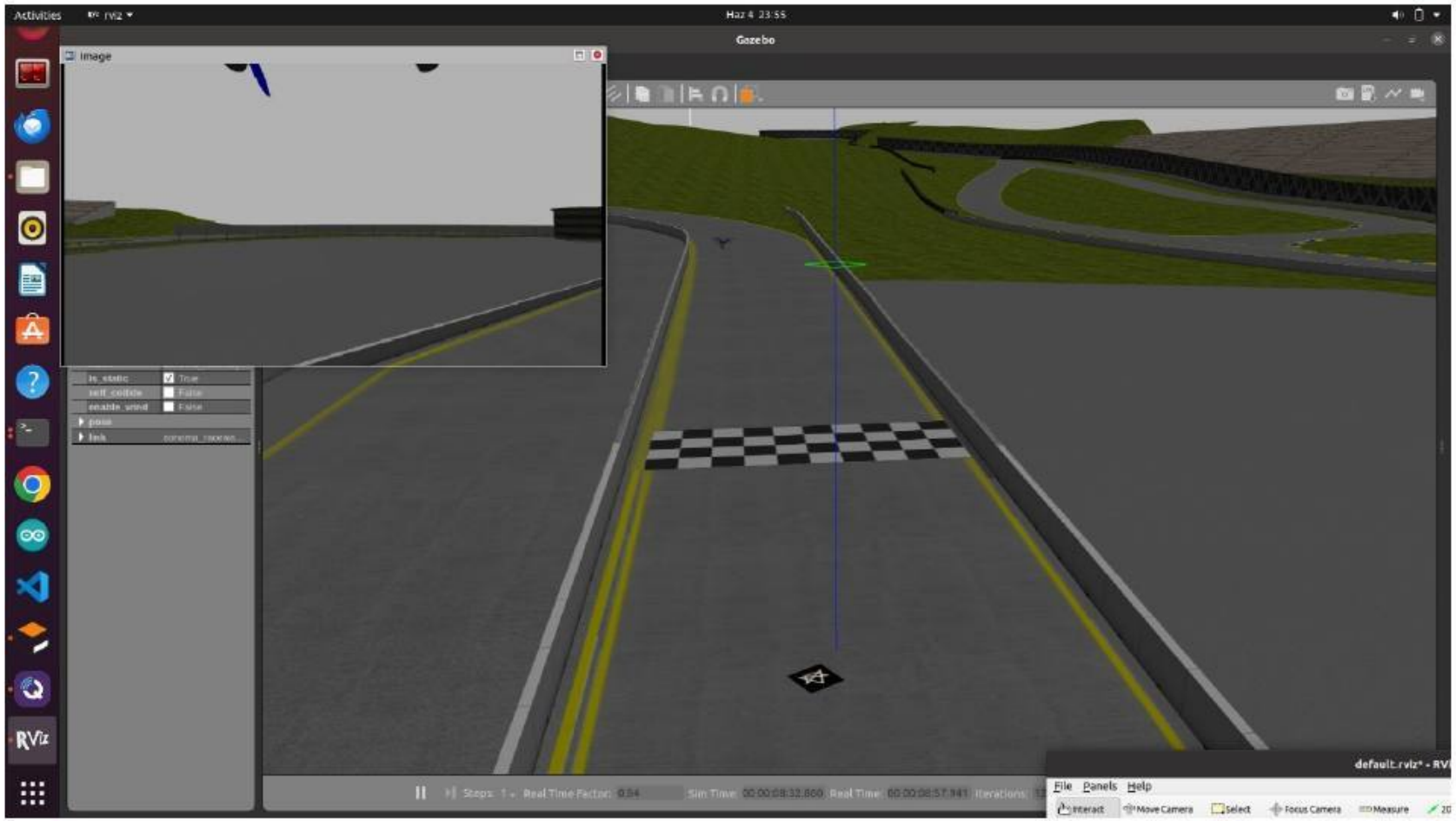


Figure 1:A visual from a mission scenario in the Gazebo simulation environment.

YILDIZ TEKNİK ÜNİVERSİTESİ

THANK YOU FOR LISTENING...